

# **CLM/Bristol Babcock RTU Interface**

**LK12-110**



**Communication  
Link Module**

# ***CLM/Bristol Babcock RTU Interface***

**LK12-110**

**1/09**

# Copyright, Notices, and Trademarks

---

Printed in U.S.A. – © Copyright 2009 by Honeywell International Inc.

Revision 09– January 2009

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customer.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

**TotalPlant** Solution ( TPS ) System is a U. S. registered trademark of Honeywell Inc. TPS is the evolution of TDC 3000x.

Local Control Network, and LCN are trademarks of Honeywell Inc.

**TotalPlant** is a U. S. registered trademark of Honeywell Inc.

ACCOL and ACCOL II are trademarks of Bristol Babcock.

IBM AT and PC DOS are trademarks of IBM Corporation.

Information Mapping is a trademark of Information Mapping Inc.

MagneW and TDC 3000<sup>X</sup> are trademarks of Honeywell Inc.

Microsoft and MS DOS are trademarks of Microsoft Corporation.

Honeywell International  
Process Solutions  
2500 West Union Hills  
Phoenix, AZ 85027  
1-800 343-0228

# About This Publication

---

This document is the interface manual for the TPS Communication Link Module (AM/CLM) and the Bristol Babcock Network 3000 products which include:

- DPC 3330
- DPC 3335

In this document, these Bristol Babcock products are collectively referred to as Remote Terminal Units (RTUs).

The CLM provides a standard communications platform, which allows the TPS system to interface with a wide variety of devices/networks. With this interface, the TPS station operator has access to points located on a non-Honeywell network. Access to the RTU is transparent, so the RTU points appear to be directly connected to the Local Control Network (LCN) on the TPS system.

The complete CLM functionality is segmented into the following four logical layers:

- Process Application
- CLM Platform
- Network Interface
- Communication Protocol

The bottom three logical layers provide the communications foundation for the top layer or Process Application layer. The Process Application layer is the actual user application and is not considered part of the CLM Platform.

This document is primarily concerned with the bottom two logical layers: Network Interface and Communication Protocol, with a specific orientation to interfacing Bristol Babcock Network 3000 products to the TPS. This document supplements the *CLM Platform Implementation Manual*. It covers the CLM Platform logical layer and the portion of the Network Interface logical layer that is not oriented to a specific non-Honeywell network.

This document is primarily intended for the engineer who will configure the interface between the TPS system and the Bristol Babcock Network 3000. The document also includes information that supplements the *CLM Platform Implementation Manual* with specific details regarding Bristol Babcock's network and products.

# Support and Other Contacts

| Location                                                                                                                                                                                                              | Organization                      | Address                                                                                                                                                                                 | Phone/Facsimile                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| United States and Canada                                                                                                                                                                                              | Honeywell Solution Support Center | Honeywell TAC,<br>MS P13, 2500 West<br>Union Hills Drive,<br>Phoenix, AZ, 85027                                                                                                         | Phone:<br>1-800 822-7673<br><br>602- 313-5558<br>(In Arizona)<br><br>Facsimile:<br>(602) 313-3293                                          |
| <b>Note:</b> Calls are answered by dispatcher between 6:00 A.M. and 4:00 P.M. Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour. |                                   |                                                                                                                                                                                         |                                                                                                                                            |
| Europe                                                                                                                                                                                                                | Honeywell TAC-EMEA                | TAC-BE02<br>Hermes Plaza<br>Hermeslaan,<br>1HB-1831 Diegem,<br>Belgium                                                                                                                  | Phone:<br>+32-2-728-2732<br><br>Facsimile:<br>+32-2-728-2696                                                                               |
| Pacific                                                                                                                                                                                                               | Honeywell Global TAC – Pacific    | Honeywell Limited<br>Australia<br>5 Kitchener Way<br>Burswood 6100,<br>Western Australia<br><br>Email:<br>GTAC@honeywell.com                                                            | Phone:<br>1300-300-4822<br>(toll free within Australia)<br><br>+61-8-9362-9559<br>(outside Australia)<br><br>Facsimile:<br>+61-8-9362-9564 |
| India                                                                                                                                                                                                                 | Honeywell Global TAC – India      | Honeywell Automation<br>India Ltd.<br>56 and 57, Hadapsar<br>Industrial Estate<br>Hadapsar,<br>Pune –411 013,<br>India<br><br>Email:<br>Global-TAC-<br>India@honeywell.com              | Phone:<br>+91-20- 66039400<br><br>Facsimile:<br>+91-20- 66039800                                                                           |
| Korea                                                                                                                                                                                                                 | Honeywell Global TAC – Korea      | Honeywell Co., Ltd<br>17F, Kikje Center B/D,<br>191, Hangangro-2Ga<br>Yongsan-gu, Seoul, 140-<br>702, Korea<br><br>Email:<br>Global-TAC-<br>Korea@honeywell.com                         | Phone:<br>+82-2-799-6317<br><br>+82-11-9227-6324<br><br>Facsimile:<br>+82-2-792-9015                                                       |
| People's Republic of China                                                                                                                                                                                            | Honeywell Global TAC – China      | Honeywell (China) Co.,<br>Ltd<br>33/F, Tower A, City<br>Center, 100 Zunyi Rd.,<br>Shanghai 200051,<br>People's Republic of<br>China<br><br>Email:<br>Global-TAC-<br>China@honeywell.com | Phone:<br>+86- 21-52574568                                                                                                                 |

| Location         | Organization                                                                                                  | Address                                                                                                                                                                                    | Phone/Facsimile                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Singapore        | Global TAC – South East Asia                                                                                  | Honeywell Private Limited<br>Honeywell Building 17,<br>Changi Business Park<br>Central 1,<br>Singapore 486073<br><br>Email:<br>GTAC-SEA@honeywell.com                                      | Phone:<br>+65-6580-3500<br><br>Facsimile:<br>+65-6580-3501<br><br>+65-6445-3033 |
| Taiwan           | Global TAC – Taiwan                                                                                           | Honeywell Taiwan Ltd.<br>17F-1, No. 260,<br>Jhongshan 2nd Road.<br>Cianjhen District<br>Kaohsiung, Taiwan,<br>ROC<br><br>Email:<br>Global-TAC-Taiwan@honeywell.com                         | Phone:<br>+886- 7- 536 2567<br><br>Facsimile:<br>+886-7-536 2039                |
| Japan            | Global TAC – Japan                                                                                            | Honeywell Japan Inc.<br>New Pier Takeshiba,<br>South Tower Building,<br>20th Floor, 1-16-1<br>Kaigan, Minato-ku,<br>Tokyo 105-0022, Japan<br><br>Email: Global-TAC-JapanJA25@honeywell.com | Phone:<br>+81-3-6730-7160<br><br>Facsimile:<br>+81-3-6730-7228                  |
| Elsewhere        | Call your nearest Honeywell office.                                                                           |                                                                                                                                                                                            |                                                                                 |
| World Wide Web   | Honeywell Solution Support Online: <a href="http://www.honeywell.com/ps">http://www.honeywell.com/ps</a>      |                                                                                                                                                                                            |                                                                                 |
| Training Classes | Honeywell Automation College: <a href="http://www.automationcollege.com">http://www.automationcollege.com</a> |                                                                                                                                                                                            |                                                                                 |



# Table of Contents

---

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| <b>Section 1 – Introduction.....</b>                                    | <b>1</b>  |
| 1.1    Overview .....                                                   | 1         |
| <b>Section 2 – Communication Link Module.....</b>                       | <b>3</b>  |
| 2.1    Communication Link Module Logical Structure.....                 | 3         |
| <b>Section 3 – Bristol Babcock RTU Data Structures .....</b>            | <b>5</b>  |
| 3.1    Introduction .....                                               | 5         |
| 3.2    ACCOL Modules .....                                              | 5         |
| 3.3    ACCOL Module Listing.....                                        | 8         |
| 3.4    Building CLM/Bristol Babcock Interface Structures .....          | 11        |
| <b>3.4.1    Building Network Data Structures.....</b>                   | <b>12</b> |
| <b>3.4.2    Building Point Interface Structures .....</b>               | <b>13</b> |
| 3.5    Files Supplied with the Bristol Babcock RTU Interface .....      | 14        |
| 3.6    Supplied Templates for Bristol Babcock RTU Interface.....        | 15        |
| <b>Section 4 – Bristol Babcock Software Installation to TPS .....</b>   | <b>17</b> |
| 4.1    Overview .....                                                   | 17        |
| 4.2    Install RTU Network Interface Layer - Setup .....                | 17        |
| 4.3    Execute the Install Procedure to HM.....                         | 18        |
| 4.4    Compile Pictures .....                                           | 19        |
| 4.5    Configure the CLMI /Port .....                                   | 20        |
| 4.6    Adding Additional Bristol Babcock RTU Network .....              | 22        |
| <b>Section 5 – Bristol Babcock Network 3000 Interface Design.....</b>   | <b>23</b> |
| 5.1    Overview .....                                                   | 23        |
| 5.2    Network Design and Configuration.....                            | 23        |
| <b>5.2.1    NDT Template Definition Files for Bristol Babcock .....</b> | <b>25</b> |
| <b>5.2.2    Port NDT Builder Input File.....</b>                        | <b>26</b> |
| <b>5.2.3    Device ( RTU ) NDT Builder Input File.....</b>              | <b>30</b> |
| 5.3    NDT Status Displays .....                                        | 32        |
| 5.4    PITG and PIT Template Definition Files .....                     | 33        |
| <b>Section 6 –RTU Image Point Interface Design .....</b>                | <b>35</b> |
| 6.1    Design and Configure.....                                        | 35        |
| 6.2    CLMI Supported RTU Conversions.....                              | 37        |

|                                                          |                                                                         |            |
|----------------------------------------------------------|-------------------------------------------------------------------------|------------|
| 6.2.1                                                    | Check Before Operate (CBO) .....                                        | 39         |
| 6.3                                                      | Image Point Interface Details .....                                     | 45         |
| 6.3.1                                                    | AGA3 Flow Point (AM Regulatory Point Type) .....                        | 46         |
| 6.3.2                                                    | AGA5 Flow Point (AM Regulatory Point Type) .....                        | 49         |
| 6.3.3                                                    | AGA7 Flow Point (AM Regulatory Point Type) .....                        | 52         |
| 6.3.4                                                    | ANIN, High- and Low-Level Analog Input (AM Regulatory Point Type) ..... | 55         |
| 6.3.5                                                    | ANOUT, Analog Output (AM Regulatory Point Type) .....                   | 57         |
| 6.3.6                                                    | ARRAYI, ARRAYO, Array Input/Output .....                                | 60         |
| 6.3.7                                                    | HSCount/LSCount, Counter (Regulatory Point Type) .....                  | 62         |
| 6.3.8                                                    | Digital Composite Points .....                                          | 64         |
| 6.3.8.1                                                  | Digital Composite Points (3-State) (AM Switch Point Type) .....         | 65         |
| 6.3.8.2                                                  | Digital Composite Point (4-State) (AM Switch Point Type) .....          | 66         |
| 6.3.8.3                                                  | Digital Composite Point (5-State) (AM Switch Point Type) .....          | 68         |
| 6.3.9                                                    | Digital Input Points .....                                              | 70         |
| 6.3.9.1                                                  | DIN3, Digital Input 3-State Builder Input .....                         | 70         |
| 6.3.9.2                                                  | DIN4, Digital Input 4-State Builder Input .....                         | 70         |
| 6.3.10                                                   | FLAG, Flag Point (AM Flag Point Type) .....                             | 71         |
| 6.3.11                                                   | FPV, Supercompressibility Point (AM Regulatory Point Type) .....        | 72         |
| 6.3.12                                                   | HWSTI, Smart Transmitter Point (AM Regulatory Point Type) .....         | 74         |
| 6.3.13                                                   | Numeric Point (AM Numeric Point Type) .....                             | 79         |
| 6.3.14                                                   | PID3TM, PID Controller Point (AM Regulatory Point Type) .....           | 80         |
| <b>Section 7 – Smart Transmitter Configuration .....</b> |                                                                         | <b>87</b>  |
| 7.1                                                      | Honeywell Smartline Transmitter Network Interface .....                 | 87         |
| 7.2                                                      | Bring Up the Smart Transmitter Configuration Display .....              | 87         |
| 7.3                                                      | Smart Transmitter Configuration Display .....                           | 88         |
| 7.4                                                      | Smart Transmitter Display After DECONF Selection .....                  | 90         |
| 7.5                                                      | Configuration Data of Smart Transmitter Display .....                   | 91         |
| 7.6                                                      | Parameter Description of Smart Transmitter Display .....                | 93         |
| <b>Section 8 – Terminology and Definitions .....</b>     |                                                                         | <b>99</b>  |
| 8.1                                                      | Terms Used in This Document .....                                       | 99         |
| 8.2                                                      | CLM/Bristol Babcock RTU Interface Package .....                         | 99         |
| 8.3                                                      | Other Reference Documents .....                                         | 99         |
| <b>Appendix – UCLM: CCP AND UCIM Redesigned .....</b>    |                                                                         | <b>101</b> |

# Acronyms

---

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| ACCOL     | Advanced Command & Control Operator Language (Bristol Babcock RTU) |
| AGA       | American Gas Association                                           |
| AI        | Analog Input                                                       |
| AM        | TPS Application Module                                             |
| AO        | Analog Output                                                      |
| BIF       | CLM Builder Input File                                             |
| BSAP      | Bristol Standard Asynchronous Protocol                             |
| CBO       | Check Before Operate                                               |
| CDS       | Custom Data Segment                                                |
| CL        | Control Language                                                   |
| CLM       | Communication Link Module                                          |
| CLMI      | Communication Link Module Interface                                |
| DE        | Digital Enhanced Communications (Smart Transmitter Protocol)       |
| DEB       | Data Entity Builder                                                |
| DI        | Digital Input                                                      |
| DO        | Digital Output                                                     |
| DPC       | Distributed Process Controller                                     |
| DPC 3330  | A DPC package for mounting on a wall, or in a NEMA 4 enclosure     |
| DPC 3335  | A DPC package for rack mounting                                    |
| 7HM       | History Module                                                     |
| LCN       | Local Control Network                                              |
| NaN       | A Floating Point Value Which is Not a Number                       |
| NCF       | Network Configuration File                                         |
| NDT       | Network Definition Table                                           |
| NIM       | Network Interface Module                                           |
| OP        | Output                                                             |
| PID       | Proportional / Integral / Derivative                               |
| PIT       | Point Interface Table                                              |
| PITE      | Point Interface Table Entry                                        |
| PITG      | Point Interface Table Group                                        |
| PLC       | Programmable Logic Controller                                      |
| PM        | Process Manager                                                    |
| PPS       | Point Process Special                                              |
| PROM      | Programmable Read Only Memory                                      |
| PV        | Process Variable                                                   |
| RBE       | Report by Exception                                                |
| RDB       | Remote Database                                                    |
| RSN       | RBE Request Serial Number                                          |
| RTU       | Remote Terminal Unit                                               |
| SFC       | Smart Field Communicator                                           |
| SP        | Set Point                                                          |
| SPT       | Smart Pressure Transmitter                                         |
| STT       | Smart Temperature Transmitter                                      |
| SV        | Secondary Variable                                                 |
| TDF       | CLM Template Definition File                                       |
| TPS       | TotalPlant Solution or TPS system                                  |
| US or GUS | TPS Operator / Engineer Station                                    |



## Section 1 – Introduction

### 1.1 Overview

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>           | This manual describes how to interface the Bristol Babcock Remote Terminal Unit (RTU) network to the TPS Local Control Network (LCN). The RTU network is called the Bristol Babcock Network 3000. Three different types of RTUs may be used, the DPC 3330, the DPC 3335, and the DPC 3310.                                                                                                                                         |
| <b>What this manual does</b>  | This manual documents the network interface that defines the Bristol Babcock Network 3000 and the devices connected to it. It includes the image points used to reflect the data on the Bristol Babcock Network 3000, and data conversion information. The document also includes information that supplements the <i>CLM Platform Implementation Manual</i> with specific details regarding Bristol Babcock network and products. |
| <b>What it does not do</b>    | This manual does not document how to write specific process applications using the Bristol Babcock RTU. Please refer to the <i>CLM Platform Implementation Manual</i> for any additional information that may be required to complete a CLM/Bristol Babcock Interface implementation.                                                                                                                                              |
| <b>Intended audience</b>      | This document is primarily intended for the Network Interface Design Engineer who has responsibility for definition of a Bristol Babcock Network 3000 and its process data for use by the TPS Operator and the Process Engineer.                                                                                                                                                                                                   |
| <b>Prerequisite knowledge</b> | Since this manual supplements the <i>CLM Platform Implementation Manual</i> , you should become familiar with the CLM Platform manual first, especially the following sections: <ul style="list-style-type: none"><li>• Defining Interface Structure Templates</li><li>• Building CLM Interface Structures</li><li>• Network Interface Design Considerations</li></ul>                                                             |
| <b>RTU link with TPS</b>      | The RTU communicates with the TPS through the Communication Link Module (CLM) - Figure 1-1.                                                                                                                                                                                                                                                                                                                                        |

*Continued on next page*

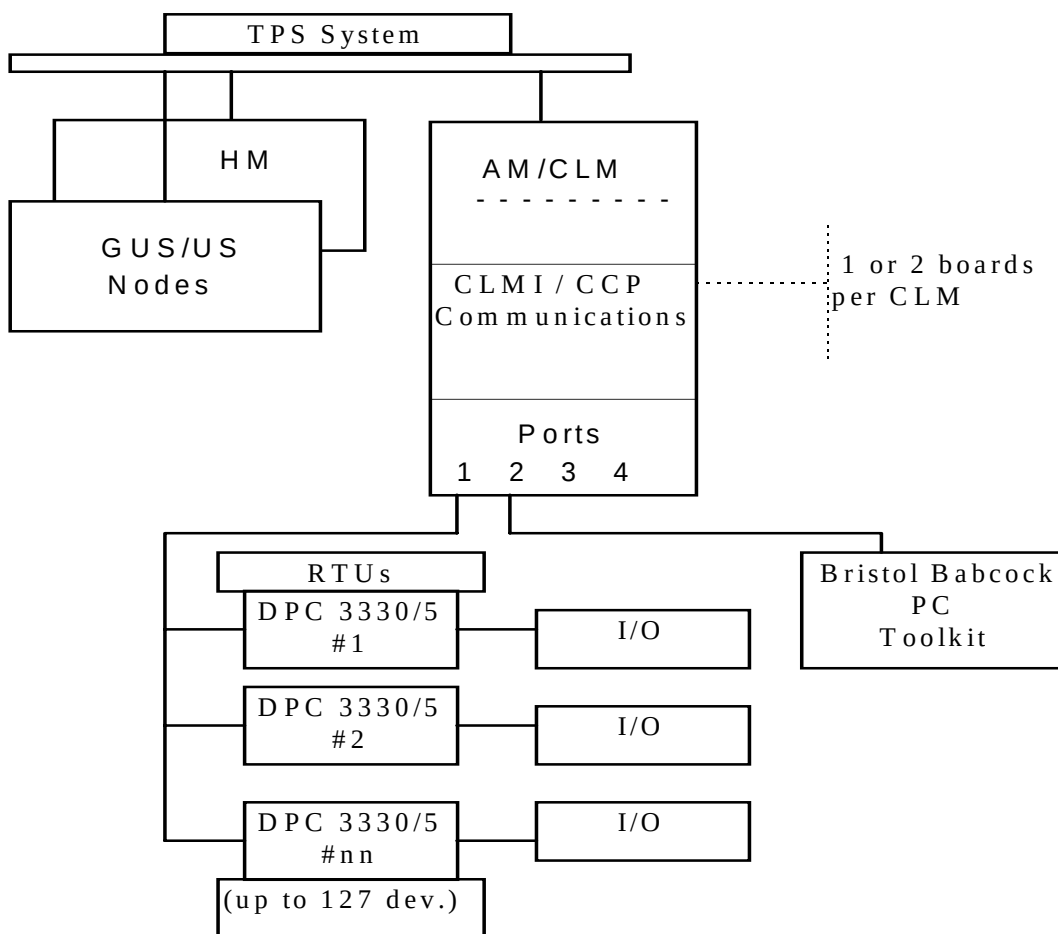
## 1.1 Overview, Continued

---

|                                                       |                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLM is a modified AM</b>                           | The CLM contains a maximum of two Communication Link Module Interface (CLMI or CCP) boards. Each of these boards has four communication ports. The RTU (or the Bristol Babcock Network) is physically connected through these ports.                                     |
| <b>AM point defines image points</b>                  | The CLM interface provides the TPS Station operator access to signals located on the Bristol Babcock Network 3000. The CLM uses image points that represent Advanced Communication and Control Language (ACCOL) modules in the Bristol Babcock Network 3000.             |
| <b>ACCOL module parameter changes by TPS operator</b> | The TPS Station operator can monitor and change selected operating parameters in the ACCOL module by changing the associated parameters in the CLM image point. Figure 1-1 shows an overview of the interface between Bristol Babcock's Network 3000 and the TPS system. |

---

Figure 1-1 CLM Interface to Bristol Babcock's Network 3000



## Section 2 – Communication Link Module

### 2.1 Communication Link Module Logical Structure

---

#### Introduction

The Communication Link Module consists of four logical layers. Refer to Section 2.3 in the *CLM Platform Implementation Manual* for further information on CLM logical structure. The CLM interface supports a user application that is actually controlling a process that includes control points on the Bristol Babcock Network 3000.

---



## Section 3 – Bristol Babcock RTU Data Structures

### 3.1 Introduction

---

#### Introduction

Bristol Babcock's Advanced Communications and Control Oriented Language (ACCOL) is a symbolic language whose building blocks are ACCOL Modules and module signals.

---

### 3.2 ACCOL Modules

---

#### ACCOL module

Each ACCOL Module:

- Performs a specific calculation or communication function
  - Emulates a hardware function
  - Is named after the function it emulates
    - ANIN for Analog Input
    - TPS Counterpart is the data point algorithm
- 

#### ACCOL module signal

Each ACCOL Module signal is:

- A data value
  - One of the following types:
    - Analog
      - High
      - Low
      - High high
      - Low low
    - Logical
      - Logical with Report by Exception
      - Logical with alarms
        - True state
        - False state
        - Change of state
        - Return to normal
    - String
  - The TPS Counterpart of the point parameter
- 

*Continued on next page*

## 3.2 ACCOL Modules, Continued

### Report by Exception

Report by Exception (RBE) provides an efficient way for the CLM to acquire data from the RTU. In this implementation of RBE the CLM polls the RTU for data and the RTU reports only the data that changes between the polled reports. To reduce the amount of RBE data that is reported, each RBE signal is configured with report deadbands that permit only significant changes to be reported. In the RTU, signals should not be configured for both alarm and RBE. RBE provides a more efficient reporting method than RTU alarm reports; however, RBE will only report changes at the scan rate of the RBE task in the RTU.

Signals that are configured for RBE reporting are only reported when a change exceeds the configured limit. RBE signals should not be configured as alarms. RBE parameter configuration and deadband in the RTU

are not visible to the Communication Link Module. This method eliminates the request/response interface between the CLM and the RTU, making data collection more efficient.

### RBE Configuration

#### ATTENTION

To configure the CLM for RBE reporting:

In the PITE Template Definition Files:

- Set the ECP bit to **Y** for the target input parameter.

In the B\_DEVICE NDT Template Definition File:

- Set RBE to **Y**
- Set RBE\_SLC. This value is used to break the scan process into an equal number of parts. To perform the entire scan at one time, set the value to 1. Any greater value establishes multiple scans.
- Set RBE\_SCN to the number in tenths of a second after which you want the RTU to look for data changes to the RBE signals.

*Continued on next page*

## 3.2 ACCOL Modules, Continued

### ACCOL Configuration

#### ATTENTION

When you configure the ACCOL in the RTU, consider the following topics:

In the ACCOL load file:

- Set the RBE option to **YES** in the target parameter's signal menu and configure the deadband value.
- Include the RBE module as a task in the ACCOL download and configure the following signals as indicated:
  - Set STOP\_XMIT\_COUNT to 0 which means no ACKs will be transmitted.
  - Set MODE to 0 which means wait for initialization.
  - Set FORMAT to 1 which means Short Format is used in the reports.
  - SCANTIME and SCANSLICE are overwritten by the values installed in RBE\_SCN and RBE\_SLC in the CLM device template file.
- Board Type and Diagnostics

The following are required or the RTU will not synchronize:

- Create a logical array of 12 rows by 16 columns.
- Set the system signal #DIAG.002. equal to the created logical array number.
- Set the system signal #DIAG.001. to manual enable.
- Set the system signal #PWRUP.000. to manual enable.

### RTU alarms with CLM

Although the RTU's alarm configuration and limits are not visible to the CLM, RTU alarms can be interfaced with the CLM. The following is a summary of RTU alarms interfaced with a CLM.

- RTU alarms are NOT directly induced into the TPS alarm system.
- RTU alarm reports are used as unsolicited input to the CLM.
- The value of the signal in alarm is stored to the AM image point.
- If the image point's alarm limits match those of the signal, the alarm is propagated into the TPS alarm system.
- Alarm reports are also used as a "report by exception" mechanism for logical signals.

### RTU signals with CLM

Signals in the RTU may include signal enable/inhibit states for alarms, control stores, and manual stores. Each state is independent of the other states. The following are the settings for control stores.

Settings for Control Signal States

| Control | State of Signal                                        |
|---------|--------------------------------------------------------|
| Enable  | RTU/ACCOL or CLM stores permitted                      |
| Inhibit | CLM and operator stores permitted, RTU/ACCOL inhibited |

## 3.3 ACCOL Module Listing

### Introduction

ACCOL Modules perform functions such as process control, measurement, calculation, and timing. The following summarizes the application and function of each module. PITE templates are supplied for those modules with an asterisk.

### Module application and function

#### ACCOL Module's Application and Function Summary

| Module Name    | Description                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| AGA3 *         | Gas volume flow through an orifice plate (AGA3 method)                                                                     |
| AGA5 *         | Gas energy from volume, no calorimeter signal (AGA5 method)                                                                |
| AGA7 *         | Gas volume flow measured by turbine flowmeter positive displacement meter or vortex shedding meter (AGA7 method)           |
| AGA8           | Compute base, flowing, and super compressibility of natural gas mixture using component mole percent data (AGA8 method)    |
| ANIN *         | Analog input module                                                                                                        |
| ANOUT *        | Analog output module                                                                                                       |
| AVERAGER       | Time average and integrate an analog or digital input signal                                                               |
| CALCULATOR     | Perform analog and logical equations                                                                                       |
| CBO            | Check before operate module                                                                                                |
| CHARACTERIZE   | Alternate gas characterization method (refer to AGA8 above)                                                                |
| COMMAND        | Flexible start-up control for a wide range of equipment such as motors, valves, or pumps (similar to HG digital composite) |
| COMPARATOR     | Compare two analog signals, providing three logical output signals                                                         |
| DIFFERENTIATOR | Differentiation of an analog signal as a function of time                                                                  |
| DIGIN *        | Digital input module                                                                                                       |
| DIGOUT         | Digital output module                                                                                                      |
| ENCODE         | General purpose multifunction encode                                                                                       |
| FPV *          | Compute natural gas supercompressibility factor ( $F_{pv}$ ) by AGA Report Number NX-19                                    |
| FUNCTION       | Used to look up values in an analog data array                                                                             |

*Continued on next page*

### 3.3 ACCOL Module Listing, Continued

#### Module application and function

#### ACCOL Module's Application and Function Summary, Continued

| Module Name | Description                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------|
| HSANIN      | High-speed analog input module                                                                      |
| HSCOUNT *   | High-speed counter                                                                                  |
| HWSTI*      | Analog input through Digital Enhanced (DE) protocol for Honeywell Smartline transmitter             |
| INTEGRATOR  | Integration of an analog signal as a function of time                                               |
| LEAD/LAG    | Controlled delay effect                                                                             |
| LLANIN *    | Low-level analog input module                                                                       |
| LSCOUNT *   | Low-speed counter                                                                                   |
| PDM         | Pulse duration module                                                                               |
| PDO         | Pulse duration output module                                                                        |
| PID3TERM *  | 3-mode process control algorithm                                                                    |
| RANIN       | Remote analog input module                                                                          |
| RANOUT      | Remote analog output module                                                                         |
| RBE         | Report by exception module                                                                          |
| RDIGIN      | Remote digital input module                                                                         |
| RDIGOUT     | Remote digital output module                                                                        |
| RHSCOUNT    | Remote high-speed counter                                                                           |
| RLLANIN     | Remote low-level analog input module                                                                |
| RLSCOUNT    | Remote low-speed counter                                                                            |
| RPDM        | Remote pulse duration output module                                                                 |
| RPDO        | Remote pulse duration output module                                                                 |
| SCHEDULER   | Equalizes the elapsed running time of a number of external devices                                  |
| SEQUENCER   | For applications where a number of operations must be performed on a repeatable or sequential basis |
| STEPPER     | Performs a series of sequential operations on a number of outputs                                   |
| STORAGE     | Essentially a track and hold device performing storage and retrieval functions used for history     |
| TIMER       | Two timed outputs for controlling events within the system                                          |
| TOT/TRND    | Accepts an analog or logical input signal and provides totalized outputs and a trend output         |
| WATCHDOG    | Times an operation within or completion of a task, and performs a digital output (DO) upon timeout  |

*Continued on next page*

### 3.3 ACCOL Module Listing Continued

#### Special feature or nonprocess modules

#### Process Module's Application and Function Summary

| Module Name | Description                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| AUDIT       | Alarm and significant event audit trail module to local ASCII device only                                                      |
| CIM         | TANO Communication Interface Module (CIM); RTU becomes a slave node in a TANO network                                          |
| CNGMASTER   | Columbia Natural Gas Network Master                                                                                            |
| CNGSLAVE    | Columbia Natural Gas Network Slave                                                                                             |
| CUSTOM      | Convert signal data into a communications format accepted by other PLC type networks                                           |
| DEMUX       | For use with a MUX module in a communications application                                                                      |
| KEYBOARD    | Operates the optional display/keypad                                                                                           |
| LOGGER      | Communication with an external ASCII logger device                                                                             |
| MASTER      | Master for peer-to-peer communications between nodes in the network                                                            |
| MUX         | Multiplexer module, analogous to a rotary switch                                                                               |
| PORTSTATUS  | On-line statistics for designated ports                                                                                        |
| REDUNDANCY  | On-line statistics for designated ports                                                                                        |
| RIOSTATS    | Monitor communication and diagnostic failures from remote I/O racks (RIOR)                                                     |
| SLAVE       | Slave for peer-to-peer communications between nodes in the network                                                             |
| SYSTEM      | Used to access task signals and display task errors                                                                            |
| SYSTEM0     | A list of system signals identifying such parameters as errors, limits, time, date, diagnostics, line, link, and node failures |

## 3.4 Building CLM/Bristol Babcock Interface Structures

---

### Introduction

All of the CLM/Bristol Babcock RTU Interface structures are defined using the CLM Configuration Builder which is part of the CLM Platform software. These interface structures include the following:

- Network Definition Table (NDT)
- Point Interface Table Group (PITG)
- Point Interface Table (PIT)
- Point Interface Table Entry (PITE)

A complete description of the CLM Configuration Builder and the build process is included in the *CLM Platform Implementation Manual*. However, this section includes a summary of the steps required to complete the definitions of the Bristol Babcock Interface structures.

---

### Configuration builder

The CLM Configuration Builder (referred to as the Builder) will accept an input definition either from a file or on-line from the TPS Station. The input file mode requires a Builder Input File (BIF) for each interface structure being defined.

---

### BIF validation with TDF

During the actual interface structure build process, the Builder validates each BIF input file using a predefined Template Definition File (TDF) for the specific interface structure type. Therefore, prior to the actual build process, the TDFs that will be accessed by the Builder must be available in the History Module (HM). Refer to Sections 5 and 6 for additional information on the TDFs.

---

### BIF files created before actual build process

In preparation for the actual build process, each BIF file may be created in advance using the TPS file editor or another text file editor. The BIF files are then transferred to the HM directory, NET>&CUS, for use by the Builder during the actual build process. The following steps describe the creation of the BIF files required for the CLM/Bristol Babcock Interface.

---

## 3.4.1 Building Network Data Structures

### Define NDT structure

#### Define and Generate NDT Network Interface Structures

| Step      | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|------|-----------------------------------|------|----------------------------------------|-----|---------------------------------------------------|----|--------------------------------|
| 1         | <p>Identify each addressable component in the network.</p> <p>The number of addressable network components will vary from site to site; however, all networks will have the following types.</p> <table><tr><th>Component</th><th>Description</th></tr><tr><td>CLMI</td><td>CLMI or CCP Board; 1 or 2 per CLM</td></tr><tr><td>PORT</td><td>1 for each Bristol Network 3000 and PC</td></tr><tr><td>RTU</td><td>1 for each Bristol Babcock RTU (DPC 3330 or 3335)</td></tr><tr><td>PC</td><td>PC with ACCOL Toolkit software</td></tr></table>                                                                                                                                                                                  | Component | Description | CLMI | CLMI or CCP Board; 1 or 2 per CLM | PORT | 1 for each Bristol Network 3000 and PC | RTU | 1 for each Bristol Babcock RTU (DPC 3330 or 3335) | PC | PC with ACCOL Toolkit software |
| Component | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| CLMI      | CLMI or CCP Board; 1 or 2 per CLM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| PORT      | 1 for each Bristol Network 3000 and PC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| RTU       | 1 for each Bristol Babcock RTU (DPC 3330 or 3335)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| PC        | PC with ACCOL Toolkit software                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| 2         | <p>Create a Builder Input File (BIF) to include one NDT entry for each of the identified addressable network components.</p> <ul style="list-style-type: none"><li>• The BIF may be created using the TPS file editor, or other text file editor, and transferred to the HM directory NET&gt;&amp;CUS for use.</li><li>• The prototype BIFs supplied by Honeywell are listed in section 3.5.</li><li>• Sample BIF listings for the CLMI, Bristol Babcock Port, and Bristol Babcock RTU Devices are included in section 5.2.</li><li>• Copy the appropriate BIF file, and complete each definition within the BIF, to create the BIF NDT entry for each addressable network component in your Bristol Babcock network.</li></ul> |           |             |      |                                   |      |                                        |     |                                                   |    |                                |
| 3         | <p>Build the NDT on the target CLM node. This is performed using the CLM Configuration Builder.</p> <p>Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |             |      |                                   |      |                                        |     |                                                   |    |                                |

## 3.4.2 Building Point Interface Structures

### Introduction

The PITG and PIT interface structures are common across all CLM implementations. Therefore, instructions for defining BIF files for these interface structures, and prototype BIF listings, are included in the Platform manual; refer to *CLM Platform Implementation Manual*.

### Define PITE structure

#### Create Point Interface Table Entry - PITE

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Identify the specific ACCOL modules and signals in the network that require an AM image point. <ul style="list-style-type: none"><li>Each component must be identified for input into the CLM node.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2    | Create a BIF for each of the identified image points. <ul style="list-style-type: none"><li>The BIF may be created using the TPS file editor, or other text file editor, and transferred to the HM directory NET&gt;&amp;CUS for use.</li><li>The prototype BIFs supplied by Honeywell are listed in section 3.5.</li><li>Sample BIF listings for each point type are included in Section 6.</li><li>Copy the appropriate BIF file and complete each definition within the BIF to create the BIF PITE entry.</li><li>Each PITE entry must be input to the CLM Configuration Builder to generate the PITE interface structure required for each image point in your Bristol Babcock network.</li></ul> |
| 3    | Build the image points and their point interface structures on the target CLM node. This is performed using the TPS Data Entity Builder and the CLM Configuration Builder, which is installed as part of the CLM platform software. <ul style="list-style-type: none"><li>Instructions for using the TPS Data Entity Builder are detailed in the <i>TPS Data Entity Builder Manual</i>.</li><li>Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.</li></ul>                                                                                                                                                                         |

## 3.5 Files Supplied with the Bristol Babcock RTU Interface

| HM NET | Directories | File Types         |
|--------|-------------|--------------------|
|        | &CUS        | .Z*, .S            |
|        | UCLM        | .EC, .CL, .AO, .DS |

### CLM Interface File Types and Extensions

| Extension | File Type                                  | Extension | File Type                                    |
|-----------|--------------------------------------------|-----------|----------------------------------------------|
| .AO       | Control Language Object Modules            | .ZC       | CLMI Configuration Modification Sample       |
| .CL       | Control Language - CDS                     | .ZI       | Builder Input File (BIF) Samples             |
| .DS       | Custom Display Schematics (Source)         | .ZS       | Status Display Files (CLMI/Port/device/etc.) |
| .EC       | Executable Modules -Install,Link/Unlink CL | .ZT       | Builder Template Definition Files (TDFs)     |
| .S        | CLMI Load Module                           |           | > Network Definition Table (NDT) Templates   |
|           |                                            |           | > Point Interface Table Entry (PITE)         |

#### BSAP Files supplied on cartridge with interface:

\* - File types EB, DS, and AO may be TPS Release dependent.  
 These files will be supplied for each supported release if required.  
 Media Directory - &BBC

| Install from media to HM | CLMI/CCP Load Module |
|--------------------------|----------------------|
| EC                       | S                    |
| B_INST                   | BSAP                 |

#### Network Configuration Files

| Device     | ZC     | ZT       | ZI       | ZS       |
|------------|--------|----------|----------|----------|
| CLMI       | B_CLMI |          |          |          |
| Port / PC  |        | B_PORT   | B_PORT   | B_PORT   |
| RTU Device |        | B_DEVICE | B_DEVICE | B_DEVICE |

#### Displays

| DS*                                 |
|-------------------------------------|
| CLMSMT00                            |
| CLMOVLY1 CLMOVLY2 CLMOVLY3 CLMOVLY4 |
| CLMOVLY5 CLMOVLY6 CLMOVLY7 CLMOVLY8 |

#### Point Image Files

| ZT       | ZI       | CL/CDS   | CL/AO*   | EC       |
|----------|----------|----------|----------|----------|
| B_AGA3   | B_AGA3   | B_AGA3   |          |          |
| B_AGA5   | B_AGA5   | B_AGA5   |          |          |
| B_AGA7   | B_AGA7   | B_AGA7   |          |          |
| B_ANIN   | B_ANIN   | B_ANIN   |          |          |
| B_ANOUT  | B_ANOUT  | B_ANOUT  | B_ANOTCL | B_ANOTCL |
| B_ARRAYI | B_ARRAYI |          |          |          |
| B_ARRAYO | B_ARRAYO |          |          |          |
| B_COUNT  | B_COUNT  | B_COUNT  |          |          |
| B_DIG3   | B_DIG3   | B_DG3CDS |          |          |
| B_DIG4   | B_DIG4   | B_DG4CDS |          |          |
| B_DIG5   | B_DIG5   | B_DG5CDS |          |          |
| B_DIN3   | B_DIN3   |          |          |          |
| B_DIN4   | B_DIN4   |          |          |          |
| B_FLAG   | B_FLAG   |          |          |          |
| B_FPV    | B_FPV    | B_FPV    |          |          |
| B_HWSTI  | B_HWSTI  | B_HWSTI  |          |          |
| B_NUMER  | B_NUMER  |          |          |          |
| B_PID3TM | B_PID3TM | B_PID3TM | B_PIDCL  | B_PIDCL  |

#### Required CLM Platform Files

| ZT   | ZI   | ZS   |
|------|------|------|
| CLMI | CLMI | CLMI |
| PITG | PITG |      |
| PIT  | PIT  |      |

#### Required CLM Platform Files by Point

| CL/CDS       | CL/CDS   | CL/AO  | EC   |
|--------------|----------|--------|------|
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
| DIG3CDS/CDS1 | CLM_CMD3 | DIG3CL | DIG3 |
| DIG4CDS/CDS1 | CLM_CMD3 | DIG4CL | DIG4 |
| DIG5CDS/CDS1 | CLM_CMD3 | DIG5CL | DIG5 |
| DIN3CDS      | CLM_CMD3 |        |      |
| DIN4CDS      | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |
|              | CLM_CMD3 |        |      |

## 3.6 Supplied Templates for Bristol Babcock RTU Interface

### Introduction

PITE Templates for Single ACCOL Modules supplied with the Bristol Babcock Network Interface.

#### PITE Template Descriptions for Single ACCOL Modules

| Template Name (ZT)          | AM Point                      | Description                                                                                                       |
|-----------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| B_AGA3                      | Regulatory                    | AGA3                                                                                                              |
| B_AGA5                      | Regulatory                    | Gas energy from volume, no calorimeter signal (AGA5 method)                                                       |
| B_AGA7                      | Regulatory                    | Gas volume flow measured by turbine flowmeter, positive displacement meter or vortex shedding meter (AGA7 method) |
| B_ANIN                      | Regulatory                    | High- and Low-Level Analog input                                                                                  |
| B_ANOUT                     | Regulatory                    | Analog output                                                                                                     |
| B_ARRAYI                    | Regulatory, Switch, or Custom | Typical Input Array Template                                                                                      |
| B_ARRAYO                    | Regulatory, Switch, or Custom | Typical Output Array Template                                                                                     |
| B_COUNT                     | Regulatory                    | High- and low-speed counters (accumulator)                                                                        |
| B_FPV                       | Regulatory                    | Supercompressibility Factor of a gas                                                                              |
| B_HWSTI                     | Regulatory                    | Smartline transmitter                                                                                             |
| B_PID3TM                    | Regulatory                    | 3 mode control algorithm                                                                                          |
| <b>Other PITE Templates</b> |                               |                                                                                                                   |
| B_DIN3                      | Switch                        | AM digital input only (3-state)                                                                                   |
| B_DIN4                      | Switch                        | AM digital input only (4-state)                                                                                   |
| B_DIG3                      | Switch                        | AM digital composite point (3-state)                                                                              |
| B_DIG4                      | Switch                        | AM digital composite point (4-state)                                                                              |
| B_DIG5                      | Switch                        | AM digital composite point (5-state)                                                                              |
| B_FLAG                      | Flag                          | AM Flag Point                                                                                                     |
| B_NUMER                     | Numeric                       | AM Numeric Point                                                                                                  |



## Section 4 – Bristol Babcock Software Installation to TPS

### 4.1 Overview

#### Introduction

This section covers only the product software installation to the TPS. You will still need to create the AM image points to reflect the Bristol Babcock network. Then, using the Builder functionality, establish the connection between the external network point values and the AM image point values. Refer to *CLM Platform Implementation Manual*, Sections 5 and 6.

### 4.2 Install RTU Network Interface Layer - Setup

#### Prior to installation procedure

Before Starting the Installation Procedure

| Step | Action                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Make a copy of the release volume.                                                                                                                                                                                           |
| 2    | Check the paths for the CL SOURCE/OBJ=NET>UCLM, CL PARAM LIST=NET>UCLM, and CL CUSTOM GDF=NET>CDSG and verify them to be correct.                                                                                            |
| 3    | Set the USER DEFLT PATH to the same path as the CL SOURCE/OBJ.<br><br>This information is used when the included CDS modules are compiled and when the executable module link/unlink directives are executed.                |
| 4    | Before the install procedure is executed, the paths must be identified and the volume of NET>&CUS> must exist. This directory was also required for installation of the CLM Platform product and should already be in place. |

Files included in the product release will be copied to your system during the install process.

## 4.3 Execute the Install Procedure to HM

---

In preparation for executing the install procedure, the following items should be reviewed:

- Read the Customer Release Guide, supplied with the package, for information that applies to your installation including HM sizing requirements, special instructions, and reinstallation information.
- Scan the file B\_INST.EC, provided on the supplied media, for an overview of the install process.

The install EC file generally performs the following functions:

- checks for previous installation of the package.
- determines the CLM Platform release, previously installed, for loading of the specific release dependent files to the HM. Some files to be loaded are independent of the LCN release, other files must be loaded or compiled only for a specific release.
- requires confirmation of load path ( directories ) setup - Section 4.2.
- copies required files from the supplied media to the designated HM directories.
- compiles supplied Custom Data Segments ( CDS's ) required for package operation.
- creates an .LS file of the install execution for verification of correct installation (no errors listed ) and future reference.

**Caution:** If any errors exist, they must be corrected before continuing.

On completion, the package will be resident on the destination HM and installation procedures may continue as described in the following sections.

In the Command function, within the Engineering Personality, enter the following command to begin the install procedure:

**EC \$Fn>&BBC>B\_INST.EC \$Fn**

**where n = the install media device identification (for example, \$F1)**

Follow the prompts output by the install execution and enter the required items to continue through the procedure.

---

## 4.4 Compile Pictures

---

| Step | Action                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Set the USER DEFLT PATH to the same path as your destination path for displays.                                                                                                                                                                                                                                                                                                                            |
| 2    | <p>The custom displays for the smart transmitter process will need to be compiled. To accomplish this, all of the pictures in the following list will need to be read and compiled.</p> <ul style="list-style-type: none"><li>• CLMSMT00</li><li>• CLMOVLY1</li><li>• CLMOVLY2</li><li>• CLMOVLY3</li><li>• CLMOVLY4</li><li>• CLMOVLY5</li><li>• CLMOVLY6</li><li>• CLMOVLY7</li><li>• CLMOVLY8</li></ul> |
| 3    | In the engineering personality, invoke the picture editor.                                                                                                                                                                                                                                                                                                                                                 |
| 4    | <p>Enter the command:</p> <p><b>R file name</b></p>                                                                                                                                                                                                                                                                                                                                                        |
| 5    | <p>Enter the command:</p> <p><b>COM</b></p>                                                                                                                                                                                                                                                                                                                                                                |
| 6    | Repeat steps 4 and 5 for each file listed in step 2.                                                                                                                                                                                                                                                                                                                                                       |

---

### Button execution directive

As an option, you may enter the following button execution directive to implement a button to drive the Smart Transmitter Custom Display.

```
S_ENT(ENT19G,R_ENT(16,"ENTER POINT ID"));
S_INT(INT20G,-99);
SCHEM("CLMSMT00");
```

## 4.5 Configure the CLMI /Port

---

### Bristol Babcock application layer

#### ATTENTION

The CLM Platform product does not communicate with external devices. It provides the lower level functionality to support the communication protocol layer. The communication protocol layer will include the software and other components required to communicate with the external devices. This communication protocol layer is included with the Bristol Babcock RTU Interface.

Before proceeding, the CLM Platform installation must be complete.

---

### Configuration file naming convention

Configuration information will reside in a specific file for each different CLM node on the TPS. This CLMI configuration file is an ASCII text file with the following naming convention.

- Literal **CLMI** followed by:
- Three digit CLM physical node number (example: 001-096) followed by:
- Literal **zc** file extension
- Located in the directory of NET>&CUS>

For example, NET>&CUS>CLMI024.ZC is a possible name.

The contents are configuration directives to define the CLM environment as the following example shows.

---

```
CLMI <board #>, <slot #>
LOAD <file name>
INIT <0>, <port map>
RUN <entry offset>
END
```

---

**Note:** You may use the Maintenance Personality on a TPS Station to determine the CLMI slot numbers.

---

*Continued on next page*

## 4.5 Configure the CLMI /Port, Continued

Edit your  
configuration file

Add the protocol information to the CLMI configuration file that was created during the CLM Platform software installation procedure.

| Step                     | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                          |           |                         |          |  |          |             |  |            |           |  |              |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------|-------------------------|----------|--|----------|-------------|--|------------|-----------|--|--------------|
| 1                        | Set the current pathname to NET>&CUS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                          |           |                         |          |  |          |             |  |            |           |  |              |
| 2                        | Using the text file editor, call up the text file NET>&CUS>CLMI0nn.ZC, where nn is the CLM physical node number. For example, CLMI024.ZC is the CLMI configuration file for CLM node 24.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |           |                         |          |  |          |             |  |            |           |  |              |
| 3                        | <p>Edit or add one of the following sets of directives to the CLMI0nn.ZC after the existing RUN 41100 directive:</p> <ul style="list-style-type: none"><li>• If only one CLMI port will be connected to the Bristol Babcock Network 3000 and one port to the Toolkit PC, then use the following directives:<br/>LOAD BSAP.S<br/>INIT 0,12000000</li><li>• If two CLMI ports will be connected to the Bristol Babcock Network 3000 and one port to the Toolkit PC, then use the following directives:<br/>LOAD BSAP.S<br/>INIT 0,11200000</li><li>• If three CLMI ports will be connected to the Bristol Babcock Network 3000 and one port to the Toolkit PC, then use the following directives:<br/>LOAD BSAP.S<br/>INIT 0,11120000</li></ul> <p>This should produce a CLMI configuration file for the following:</p> <table><tr><td><b><u>CLMI Board</u></b></td><td><b>or</b></td><td><b><u>CCP Board</u></b></td></tr><tr><td>CLMI 1,5</td><td></td><td>CLMI 1,5</td></tr><tr><td>LOAD CLMI.S</td><td></td><td>LOAD CCP.S</td></tr><tr><td>RUN 41100</td><td></td><td>RUN 7F001100</td></tr></table> <p>LOAD BSAP.S<br/>INIT 0,12000000 <b>OR</b> INIT 0,11200000 <b>OR</b> INIT 0,11120000<br/>END</p> <p><b>Note:</b> In the INIT directive:<br/>1 - indicates RTU port configured<br/>2 - indicates PC port configured<br/>The PC port MUST always be the last port configured. Also, only the first four digits are used by the directive.</p> | <b><u>CLMI Board</u></b> | <b>or</b> | <b><u>CCP Board</u></b> | CLMI 1,5 |  | CLMI 1,5 | LOAD CLMI.S |  | LOAD CCP.S | RUN 41100 |  | RUN 7F001100 |
| <b><u>CLMI Board</u></b> | <b>or</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b><u>CCP Board</u></b>  |           |                         |          |  |          |             |  |            |           |  |              |
| CLMI 1,5                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CLMI 1,5                 |           |                         |          |  |          |             |  |            |           |  |              |
| LOAD CLMI.S              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | LOAD CCP.S               |           |                         |          |  |          |             |  |            |           |  |              |
| RUN 41100                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RUN 7F001100             |           |                         |          |  |          |             |  |            |           |  |              |
| 4                        | Exit the editor and save the modified CLMI configuration file back in NET>&CUS>CLMI0nn.ZC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |           |                         |          |  |          |             |  |            |           |  |              |

*Continued on next page*

## 4.5 Configure the CLMI /Port Continued

---

### **Interface layer installed**

This concludes the CLM Bristol Babcock RTU Interface Layer Installation Procedure.

You may now continue the configuration efforts required to complete the setup. The next task will be to restart the AM/CLM node. The CLM Platform and the Bristol Babcock applications software will then be loaded in the CLMI as part of the standard node load. Continue with Section 5 to complete the installation.

---

## 4.6 Adding Additional Bristol Babcock RTU Network

---

### **Adding additional Bristol Babcock applications**

If adding another Bristol Babcock application process to another CLM or the existing CLM, the CLMI configuration file for that CLM will need to be modified to include the new process. This may be as simple as adding another port to the configuration or producing a new CLMI directive file.

If adding an additional port to an existing CLM, then the appropriate CLMI directive file will need to be changed.

If adding another CLM, then the CLMI directive file for that CLM will need directives added for the Bristol Babcock application. Please refer to section 4.5.

---

## Section 5 – Bristol Babcock Network 3000 Interface Design

### 5.1 Overview

#### Introduction

The Bristol Babcock's Network 3000 Interface Design consists of the following tasks:

- Network Design and Configuration
- Point Database Design and Configuration
- Custom Interfaces

### 5.2 Network Design and Configuration

#### CLM configuration for Bristol Babcock Network 3000

The following is an outline of the steps used in the design and configuration of a CLM for the Bristol Babcock Network 3000.

Obtain and understand the following documents:

- *CLM Platform Implementation Manual*
- *ACCOL Software User's Manual*

#### Addressable components

The addressable components in the Bristol Babcock Network 3000 are:

| Component | Description                                      |
|-----------|--------------------------------------------------|
| CLMI      | CLMI Board; 1 or 2 per CLM                       |
| PORT      | CLMI RS-232 Port interface; 1 to 4 Ports per CLM |
| RTU       | Bristol Babcock RTUs (DPC 3330 and DPC 3335)     |
| PC        | PC with ACCOL Toolkit software                   |

## 5.2 Network Design and Configuration, Continued

### CLM Configuration

#### CLM Configuration for Bristol Babcock Network 3000

| Step      | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|------|-------------------------------------------|-----|------|---------------------------------|-----|-----|---------------------------------------------|-------|----|--------------------------------------|--------|
| 1         | <p>Identify the TDFs for each type of addressable component in network. The TDFs will be used in building the NDT.</p> <ul style="list-style-type: none"><li>• The CLMI TDF is detailed in the <i>CLM Platform Implementation Manual</i>.</li><li>• Specific NDT TDFs for a Port and Device are listed in 3.5.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                              |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| 2         | <p>Identify the Status Definition File (STSDF) for each addressable component type in the network. This is used by the CLM Network Status Display.</p> <ul style="list-style-type: none"><li>• The STSDF is listed in 3.5.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| 3         | <p>The TDF and STSDF files were transferred to the HM directory NET&gt;&amp;CUS in the installation process.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| 4         | <p>Identify each addressable component in the network.</p> <p>The number of addressable network components will vary from site to site; however, all networks will have the following types.</p> <table><tr><th>Component</th><th>Description</th><th>Range</th></tr><tr><td>CLMI</td><td>Communication Link Module Interface Board</td><td>1-2</td></tr><tr><td>PORT</td><td>Communication Port on the CLM I</td><td>1-4</td></tr><tr><td>RTU</td><td>One for each Bristol RTU (DPC 3330 or 3335)</td><td>0-127</td></tr><tr><td>PC</td><td>Personal Computer or Loader Terminal</td><td>0 or 1</td></tr></table>                                                                                                    | Component | Description | Range | CLMI | Communication Link Module Interface Board | 1-2 | PORT | Communication Port on the CLM I | 1-4 | RTU | One for each Bristol RTU (DPC 3330 or 3335) | 0-127 | PC | Personal Computer or Loader Terminal | 0 or 1 |
| Component | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Range     |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| CLMI      | Communication Link Module Interface Board                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1-2       |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| PORT      | Communication Port on the CLM I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1-4       |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| RTU       | One for each Bristol RTU (DPC 3330 or 3335)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0-127     |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| PC        | Personal Computer or Loader Terminal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0 or 1    |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| 5         | <p>Create a BIF to include one NDT entry for each of the identified addressable network components.</p> <ul style="list-style-type: none"><li>• The BIF may be created using the TPS file editor, or other text file editor, and transferred to the HM directory NET&gt;&amp;CUS for use.</li><li>• The prototype BIFs supplied by Honeywell are listed in 3.5.</li><li>• Sample BIF listings for the CLMI, Port, and RTU Devices are included in 5.2.1, following the TDF listings for each interface structure.</li><li>• Copy the appropriate BIF file, and complete each definition within the BIF, to create the BIF NDT entry for each addressable network component in your Bristol Babcock network.</li></ul> |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |
| 6         | <p>Build the NDT on the target CLM node. This is performed using the CLM Configuration Builder, which is installed as part of the CLM platform software.</p> <ul style="list-style-type: none"><li>• Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                    |           |             |       |      |                                           |     |      |                                 |     |     |                                             |       |    |                                      |        |

*Continued on next page*

## 5.2.1 NDT Template Definition Files for Bristol Babcock

### CLMI, Port, and Device templates

The following BIFs are used to define the CLMI, Port, and Device.

- CLMI Template File ( CLM Platform )
- B\_PORT Template File
- B\_DEVICE Template File

### CLMI/Typical BIF

```
ENTRY      = CLMI
NAME       =
A1_CLMI    =
FAILOVER   =
END
```

### Identifying CLMI boards

| Step | Field    | Action                                                                                                                                                            |
|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | NAME     | Enter a name identifier for the CLMI board component. The suggested names are <b>CLMI_1</b> for the first CLMI board and <b>CLMI_2</b> for the second CLMI board. |
| 2    | A1_CLMI  | Enter the CLMI number: 1 or 2. Enter 1 for the first CLMI board and 2 for the second CLMI board.                                                                  |
| 3    | FAILOVER | Enter a <b>Y</b> for redundancy else the default = <b>N</b> . Refer to Section 9 of the <i>CLM Platform Implementation Manual</i> .                               |

*Continued on next page*

## 5.2.2 Port NDT Builder Input File

---

### B\_PORT/BIF

B\_PORT/Typical BIF for the Bristol Babcock Port

```
ENTRY      = B_PORT
NAME       =
A1_CLMI    =
A2_PORT    =
FAILOVER   =
* (0=300 1=1200 2=2400 3=4800 4=9600 5=19200 6=38400)
BAUD       =
DBITS      =
SBITS      =
* (NONE=0, ODD=1, EVEN=2)
PARITY     =
RETRY      =
TIMEOUT    =
* Bristol polling rate, seconds
POLLRATE   =
* Number of RTUs connected to this port
NUMRTUS    =
Number of RTUs on level 1 of network
L1_RTUS    =
Number of RTUs on level 2 of network
L2_RTUS    =
Number of RTUs on level 3 of network
L3_RTUS    =
Number of RTUs on level 4 of network
L4_RTUS    =
Number of RTUs on level 5 of network
L5_RTUS    =
Number of RTUs on level 6 of network
L6_RTUS    =
END
```

*Continued on next page*

## 5.2.2 Port NDT Builder Input File Continued

### Identifying ports

#### B\_PORT definitions

| Step | Field     | Action                                                                                                                                                                                                                                                                                                                                               |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|---|-----|---|------|---|------|---|------|---|------|---|-------|
| 1    | NAME      | Enter a name identifier for the port component.<br>The suggested names are <code>PORT_1</code> for the first port and <code>PORT_2</code> for the second port.                                                                                                                                                                                       |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 2    | A1_CLMI   | Enter the CLMI number that the port is on: 1 or 2.                                                                                                                                                                                                                                                                                                   |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 3    | A2_PORT   | Enter the port number: 1, 2, 3, or 4.                                                                                                                                                                                                                                                                                                                |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 4    | FAILOVER  | Enter a <b>Y</b> if redundant else default = <b>N</b> . (Refer to Section 9 of the <i>CLM Platform Implementation Manual</i> .)                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 5    | BAUD      | Enter the baud rate code for this communication port.<br>Valid codes and their meanings are: <table><tr><th>Code</th><th>Baud Rate</th></tr><tr><td>0</td><td>300</td></tr><tr><td>1</td><td>1200</td></tr><tr><td>2</td><td>2400</td></tr><tr><td>3</td><td>4800</td></tr><tr><td>4</td><td>9600</td></tr><tr><td>5</td><td>19200</td></tr></table> | Code | Baud Rate | 0 | 300 | 1 | 1200 | 2 | 2400 | 3 | 4800 | 4 | 9600 | 5 | 19200 |
| Code | Baud Rate |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 0    | 300       |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 1    | 1200      |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 2    | 2400      |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 3    | 4800      |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 4    | 9600      |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 5    | 19200     |                                                                                                                                                                                                                                                                                                                                                      |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 6    | DBITS     | Number of data bits in a communication byte.<br>This is 8 for both the Bristol Babcock Network 3000 port and the port connected to the Toolkit PC.                                                                                                                                                                                                   |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 7    | SBITS     | Stop bits code.<br>This is 1 for both the Bristol Babcock Network 3000 port and the port connected to the Toolkit PC.                                                                                                                                                                                                                                |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 8    | PARITY    | Data parity code.<br>This is 0 for both the Bristol Babcock Network 3000 port and the port connected to the Toolkit PC.                                                                                                                                                                                                                              |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 9    | RETRY     | Enter the RETRY code.<br>This defines the number of retries that will be attempted on a communication failure before aborting the request.<br>The default is 2, which means a total of 3 attempts to communicate will be made before aborting the request.                                                                                           |      |           |   |     |   |      |   |      |   |      |   |      |   |       |
| 10   | TIMEOUT   | Request time-out period.<br>This is the base time-out period (in tenths of a second) for a no-request communication failure before the request is retried.<br>For example, 50 means 5 seconds.                                                                                                                                                       |      |           |   |     |   |      |   |      |   |      |   |      |   |       |

*Continued on next page*

## 5.2.2 Port NDT Builder Input File Continued

### B\_PORT definitions

| Step                                                                                                                              | Field                                                                                                                                                                                                                                                                                                                                                                                                                            | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |          |                                                                                                                                 |                                                                                                                                                             |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11                                                                                                                                | POLLRATE                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>This is the communication polling rate (in tenths of a second) for the Bristol Babcock Network 3000 port connection.</p> <p>This polling rate may be obtained from the Bristol Babcock PC program POLLPER.</p> <p>A poll rate of 8 seconds would be entered as 80.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                            |          |                                                                                                                                 |                                                                                                                                                             |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 12                                                                                                                                | NUMRTUS                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>This is the number of local RTUs, level 1 only, that are defined or reserved for access through this port of the CLMI. NUMRTUS defines to the CLMI how many level 1 RTUs are configured on this port. The sum of NUMRTUS for all ports will be the number of RTUs defined at level 1 of the complete RTU network.</p> <p>For example:</p> <table><tr><th>If the following configuration existed....</th><th>Then ...</th></tr><tr><td><ul style="list-style-type: none"><li>Port 1 had RTUs 1-15 connected at level 1</li><li>• Addresses 16-59 were spare.</li></ul></td><td><ul style="list-style-type: none"><li>NUMRTUS for port 1 would be at least 15 and could be up to 59 if addresses 16-59 were configured on port 1.</li></ul></td></tr><tr><td><ul style="list-style-type: none"><li>Port 2 had RTUs 60-72 connected at level 1</li><li>• Addresses 73-127 were spare.</li></ul></td><td><ul style="list-style-type: none"><li>NUMRTUS for port 2 would be at least 13 if spare addresses 16-59 were all configured on port 1 and could be up to 68 if spare addresses 73-127 were all configured on port 2.</li></ul><p><b>Note:</b> Spare Addresses 16-59 could be configured either all on port 1 or all on port 2 or some on each port. Spare addresses 73-127 could be configured either all or part on port 2.</p></td></tr></table> | If the following configuration existed.... | Then ... | <ul style="list-style-type: none"><li>Port 1 had RTUs 1-15 connected at level 1</li><li>• Addresses 16-59 were spare.</li></ul> | <ul style="list-style-type: none"><li>NUMRTUS for port 1 would be at least 15 and could be up to 59 if addresses 16-59 were configured on port 1.</li></ul> | <ul style="list-style-type: none"><li>Port 2 had RTUs 60-72 connected at level 1</li><li>• Addresses 73-127 were spare.</li></ul> | <ul style="list-style-type: none"><li>NUMRTUS for port 2 would be at least 13 if spare addresses 16-59 were all configured on port 1 and could be up to 68 if spare addresses 73-127 were all configured on port 2.</li></ul> <p><b>Note:</b> Spare Addresses 16-59 could be configured either all on port 1 or all on port 2 or some on each port. Spare addresses 73-127 could be configured either all or part on port 2.</p> |
| If the following configuration existed....                                                                                        | Then ...                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |          |                                                                                                                                 |                                                                                                                                                             |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>Port 1 had RTUs 1-15 connected at level 1</li><li>• Addresses 16-59 were spare.</li></ul>   | <ul style="list-style-type: none"><li>NUMRTUS for port 1 would be at least 15 and could be up to 59 if addresses 16-59 were configured on port 1.</li></ul>                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |          |                                                                                                                                 |                                                                                                                                                             |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>Port 2 had RTUs 60-72 connected at level 1</li><li>• Addresses 73-127 were spare.</li></ul> | <ul style="list-style-type: none"><li>NUMRTUS for port 2 would be at least 13 if spare addresses 16-59 were all configured on port 1 and could be up to 68 if spare addresses 73-127 were all configured on port 2.</li></ul> <p><b>Note:</b> Spare Addresses 16-59 could be configured either all on port 1 or all on port 2 or some on each port. Spare addresses 73-127 could be configured either all or part on port 2.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |          |                                                                                                                                 |                                                                                                                                                             |                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                  |

*Continued on next page*

## 5.2.2 Port NDT Builder Input File Continued

### B\_PORT definitions

| Step                                                            | Field                                                | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
|-----------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|---|---|-------------|---|-------------|---|--------------|---|---------------|---|---------------|---|----------------|---|
| 13                                                              | L1_RTUS                                              | <p>This is the number of local RTUs, level 1 only, that are defined for access through all ports of the CLMI. Include spares for future RTUs.</p> <p><b>Note:</b> For L1_RTUS through L6_RTUS the following RTU addressing rules apply:</p> <ul style="list-style-type: none"><li>• Entries L1_RTUS through L6_RTUS need only be defined for port 1 NDT. That definition is used for all ports.</li><li>• RTU addresses may range from 1 to 127.</li><li>• An RTU's network address consists of the local addresses of all RTUs in the communication path from level 1 to the addressed RTU. This network or global address is limited to 15 bits which further restricts the number of RTUs at all levels in the RTU network. The following table shows the number of bits required for an RTU address:</li></ul> |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
|                                                                 |                                                      | <table><tr><td>If the highest RTU address at that level in the RTU network is:</td><td>Then the number of bits required for the address is:</td></tr><tr><td>1</td><td>1</td></tr><tr><td>from 2 to 3</td><td>2</td></tr><tr><td>from 4 to 7</td><td>3</td></tr><tr><td>from 8 to 15</td><td>4</td></tr><tr><td>from 16 to 31</td><td>5</td></tr><tr><td>from 32 to 63</td><td>6</td></tr><tr><td>from 64 to 127</td><td>7</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                       | If the highest RTU address at that level in the RTU network is: | Then the number of bits required for the address is: | 1 | 1 | from 2 to 3 | 2 | from 4 to 7 | 3 | from 8 to 15 | 4 | from 16 to 31 | 5 | from 32 to 63 | 6 | from 64 to 127 | 7 |
| If the highest RTU address at that level in the RTU network is: | Then the number of bits required for the address is: |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| 1                                                               | 1                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 2 to 3                                                     | 2                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 4 to 7                                                     | 3                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 8 to 15                                                    | 4                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 16 to 31                                                   | 5                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 32 to 63                                                   | 6                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |
| from 64 to 127                                                  | 7                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                 |                                                      |   |   |             |   |             |   |              |   |               |   |               |   |                |   |

*Continued on next page*

## 5.2.2 Port NDT Builder Input File Continued

### B\_PORT definitions

|           |         |                                                                                                                                                                                                                                                            |
|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>14</b> | L2_RTUS | Highest RTU address, configured or reserved, at any level 2 segment of the RTU network.<br>Default of 0 - level 2 is not used.                                                                                                                             |
| <b>15</b> | L3_RTUS | Highest RTU address, configured or reserved, at any level 3 segment of the RTU network.<br>Default of 0 - level 3 is not used.                                                                                                                             |
| <b>16</b> | L4_RTUS | Highest RTU address, configured or reserved, at any level 4 segment of the RTU network.<br>Default of 0 - level 4 is not used.                                                                                                                             |
| <b>17</b> | L5_RTUS | Highest RTU address, configured or reserved, at any level 5 segment of the RTU network.<br>Default of 0 - level 5 is not used.                                                                                                                             |
| <b>18</b> | L6_RTUS | Highest RTU address, configured or reserved, at any level 6 segment of the RTU network.<br>Default of 0 - level 6 is not used.                                                                                                                             |
| <b>19</b> |         | Check configuration – using the table under Step 13 above, find the number of bits required for the highest address at each level (L1_RTUS through L6_RTUS). If the sum of these bits exceed 15 the configuration is not valid. Levels must be contiguous. |

## 5.2.3 Device ( RTU ) NDT Builder Input File

### B\_DEVICE/Typical BIF for the Bristol Babcock Device

```
ENTRY      = B_DEVICE
NAME       =
A1_CLMI    =
A2_PORT    =
A3_DEV     =
A4_DEV2    =
A5_DEV3    =
A6_DEV4    =
A7_DEV5    =
A8_DEV6    =
RBE        =
RBE_SCN    =
RBE_SLC    =
CBO_TIM    =
END
```

## 5.2.3 Device ( RTU ) NDT Builder Input File Continued

### Identifying RTUs

| Step | Field   | Action                                                                                                                                                                                                                                                                                                                     |
|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | NAME    | Enter a name identifier for the RTU. The suggested names are RTU_1, RTU_2, RTU_3, etc.                                                                                                                                                                                                                                     |
| 2    | A1_CLMI | Enter the CLMI number: 1 or 2.                                                                                                                                                                                                                                                                                             |
| 3    | A2_PORT | Enter the port number: 1, 2, 3, or 4.                                                                                                                                                                                                                                                                                      |
| 4    | A3_DEV  | Enter the RTU address at level 1.<br><b>Note:</b> Parameters A3_DEV through A8_DEV6 defines the network or global address of this RTU network communication path from level 1 to this RTU. The local address is the address from the node address switches on the RTU. Enter -1 for all address parameters after this RTU. |
| 5    | A4_DEV2 | Enter the RTU address at level 2 or -1 if this level does not apply to this RTU.                                                                                                                                                                                                                                           |
| 6    | A5_DEV3 | Enter the RTU address at level 3 or -1 if this level does not apply to this RTU.                                                                                                                                                                                                                                           |
| 7    | A6_DEV4 | Enter the RTU address at level 4 or -1 if this level does not apply to this RTU.                                                                                                                                                                                                                                           |
| 8    | A7_DEV5 | Enter the RTU address at level 5 or -1 if this level does not apply to this RTU.                                                                                                                                                                                                                                           |
| 9    | A8_DEV6 | Enter the RTU address at level 6 or -1 if this level does not apply to this RTU.                                                                                                                                                                                                                                           |
| 10   | RBE     | Enter <b>Y</b> if this RTU is configured for RBE, otherwise enter <b>N</b> .                                                                                                                                                                                                                                               |
| 11   | RBE_SCN | Enter the execution cycle for the exception report task in the RTU. This cycle time is entered in tenths of a second and ranges from 1 to 65535.                                                                                                                                                                           |
| 12   | RBE_SLC | Enter the number of segments each execution cycle is to be divided into by the RTU exception report task. This permits other RTU tasks to execute during the RBE scan cycle. This may be desired in RTUs with short scan cycles and have many RBE signals configured.                                                      |
| 13   | CBO_TIM | Bristol CBO request timeout in minutes.                                                                                                                                                                                                                                                                                    |

#### **WARNING**

If the RTU is not equipped for the RBE feature, DO NOT respond with **Y** in step 5. Choosing **Y** will cause the CLM RTU to remain in the SYNC state until the NDT is reconfigured for the RTU.

## 5.3 NDT Status Displays

---

### Introduction

The CLM STSDF layout is described in the *CLM Platform Implementation Manual* (CLM Status Definition File Design Section). The following gives sample display layouts for the CLM, Port, and Device.

---

### Display for CLMI

### – NO. OF UNACKNOWLEDGED FAILED DEVICES ON THE CLMI  
### – NO. OF ACKNOWLEDGED FAILED DEVICES ON THE CLMI

Average Parameters per Second

|        | INPUT | OUTPUT |
|--------|-------|--------|
| 4 Sec. | ####  | ####   |
| 5 Min. | ####  | ####   |

##### – NO. OF PPSs [CLM\_PPS, CLM\_OUT, and PPS]

---

### Display for port

### – NO. OF UNACKNOWLEDGED FAILED DEVICES ON THE PORT  
### – NO. OF ACKNOWLEDGED FAILED DEVICES ON THE PORT

##### – NUMBER OF DATA MESSAGES RECEIVED  
##### – NUMBER OF DATA MESSAGES TRANSMITTED  
##### – NUMBER OF RESPONSE TIMEOUTS  
##### – NUMBER OF CONSECUTIVE RESPONSE TIMEOUTS  
##### – NUMBER OF NAKS RECEIVED  
##### – NUMBER OF CRC ERRORS  
##### – NUMBER OF MESSAGE DISCARDED ACKS RECEIVED  
##### – NUMBER OF PROTOCOL ERRORS  
##### – NUMBER OF ALARM ELEMENTS RECEIVED  
##### – NUMBER OF RBE ELEMENTS RECEIVED  
##### – NUMBER OF RSN ERRORS

Baud Rate ##### Retry ### Timeout ### Data Bits ## Parity ##

---

*Continued on next page*

## 5.3 NDT Status Displays Continued

### Display for device

|                                       |                                     |
|---------------------------------------|-------------------------------------|
| ##### - DATA MESSAGES TRANSMITTED     | ##### - RESPONSE TIMEOUTS           |
| ##### - CONSECUTIVE RESPONSE TIMEOUTS | ##### - NAKS RECEIVED               |
| ##### - CRC ERRORS                    | ##### - MSG DISCARDED ACKS RECEIVED |
| ##### - PROTOCOL ERRORS               | ##### - ALARM ELEMENTS RECEIVED     |
| ##### - RBE ELEMENTS RECEIVED         | ##### - RSN ERRORS                  |

| Board |      |       |        | Status |      |       |        |
|-------|------|-------|--------|--------|------|-------|--------|
| No.   | Type | Major | Detail | No.    | Type | Major | Detail |
| 1     | 0    | OK    | 0      | 2      | 0    | OK    | 0      |
| 3     | 0    | OK    | 0      | 4      | 0    | OK    | 0      |
| 5     | 0    | OK    | 0      | 6      | 0    | OK    | 0      |
| 7     | 0    | OK    | 0      | 8      | 0    | OK    | 0      |
| 9     | 0    | OK    | 0      | 10     | 0    | OK    | 0      |
| 11    | 0    | OK    | 0      | 12     | 0    | OK    | 0      |

**Note:** Board types are defined in the Bristol Babcock documentation.

## 5.4 PITG and PIT Template Definition Files

### TDFs for PITG and PIT templates

The TDFs for both the PITG and PIT templates are common across all CLM implementations.

Instructions and listings for the PITG and PIT Template Definition Files are detailed in the *CLM Platform Implementation Manual*.



## Section 6 –RTU Image Point Interface Design

### 6.1 Design and Configure

#### Image point design and configuration process

---

The following is an outline of the steps used in the design and configuration of an image point interface to a Bristol Babcock Network 3000 RTU.

Obtain and understand the following documents:

- *CLM Platform Implementation Manual*
- *ACCOL User's Manual*

---

**Note:** Before designing and configuring image points, you must have completed Network Design and Configuration. This activity is described in Section 5.

---

## 6.1 Design and Configure Continued

### Design and Configure Image Point Interface

| Step | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Identify the types of ACCOL modules and signals that are to be reflected as AM image points. <ul style="list-style-type: none"><li>Section 3.6 provides a list of ACCOL modules. Modules that are supported as part of this package are identified with an asterisk.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2    | Define and create the AM image points, their CDS structures, and CL, if any. <ul style="list-style-type: none"><li>ACCOL modules (identified with an asterisk) support includes: the AM image points, their CDS structures, and CL, if any.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3    | Identify the TDFs for each type of image point. The TDFs will be used in building the PIT structures. <ul style="list-style-type: none"><li>The TDFs for the PITG and PIT are listed in the <i>CLM Platform Implementation Manual</i>.</li><li>Supported PITEs are listed in 3.5.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4    | The TDF and image point files were transferred to the HM directory NET>&CUS in the installation process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5    | Identify the specific ACCOL modules and signals in the network that require an AM image point. <ul style="list-style-type: none"><li>Each component must be identified for input into the CLM node.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 6    | Create a BIF for each of the identified image points. <ul style="list-style-type: none"><li>The BIF may be created using the TPS file editor, or other text file editor, and transferred to the HM directory NET&gt;&amp;CUS for use.</li><li>The prototype Builder Input Files supplied by Honeywell are listed in 3.5.</li><li>Sample BIF listings for each point type are included in 6.3.</li><li>Copy the appropriate BIF file, and complete each definition within the BIF, to create the BIF PITE entry.</li><li>Each PITE entry must be input to the CLM Configuration Builder, to generate the PITE interface structure required for each image point in the Bristol Babcock network.</li></ul> |
| 7    | Build the image points and their point interface structures on the target CLM node. This is performed using the TPS Data Entity Builder, and the CLM Configuration Builder, which is installed as part of the CLM platform software. <ul style="list-style-type: none"><li>Instructions for using the TPS Data Entity Builder are detailed in the <i>TPS Data Entity Builder Manual</i>.</li><li>Instructions for using the CLM Configuration Builder are detailed in the <i>CLM Platform Implementation Manual</i>.</li></ul>                                                                                                                                                                           |

## 6.2 CLMI Supported RTU Conversions

---

### Introduction

Parameter values are received from Bristol Babcock RTUs in Engineering Units. The platform parameter store routine will convert the data to the proper TPS format based on the LCN conversion code.

The RTU conversion code is the second part of the total PITE parameter conversion definition. Refer to the definition of the Pnn\_ICNV in the PITE template definition section of the *CLM Platform Implementation Manual* for details.

---

### Bristol Babcock RDBA questionable values

Bristol Babcock RDBA reports questionable values using the least significant bit of the mantissa. If the bit is set, the value is questionable. If the bit is reset, the value is good. The PITE can be configured to set questionable values to NaN.

Signal state parameters do not have an address entry. In the PITE, they always directly follow the signal to which they apply. When multiple mode parameters are required, they must be consecutive in the PITE. Only the first MODE can be an input parameter.

---

*Continued on next page*

## 6.2 CLMI Supported RTU Conversions, Continued

CLM supported RTU conversions, continued

Table 6-2 CLM Supported RTU Conversions, Continued

| Conversion Number                                                                                                                                |          | RTU Data Type                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LCN                                                                                                                                              | RTU      |                                                                                                                                                                                                                                                                                                       |
| 3                                                                                                                                                | 1        | Signal alarm enable status (signal state parameter)                                                                                                                                                                                                                                                   |
| 3                                                                                                                                                | 2        | Signal control enable status (signal state parameter)                                                                                                                                                                                                                                                 |
| 3                                                                                                                                                | 3        | Signal manual enable status (signal state parameter)                                                                                                                                                                                                                                                  |
| 2                                                                                                                                                | 4        | Analog value                                                                                                                                                                                                                                                                                          |
| 3                                                                                                                                                | 5        | Logical value                                                                                                                                                                                                                                                                                         |
| 5, max                                                                                                                                           | 5        | RTU Logical, LCN Enumeration                                                                                                                                                                                                                                                                          |
| 4                                                                                                                                                | 6,C,N    | String value (C=column, N=# of characters) C must be 1 for writes                                                                                                                                                                                                                                     |
| 4                                                                                                                                                | 7        | Units/Logical text (read only)                                                                                                                                                                                                                                                                        |
| 4                                                                                                                                                | 8        | Name text (read only, includes Base.Extension.Attribute)                                                                                                                                                                                                                                              |
| 4                                                                                                                                                | 9        | Descriptor text (read only)                                                                                                                                                                                                                                                                           |
| 4                                                                                                                                                | 10       | Logical on/off text (read only)                                                                                                                                                                                                                                                                       |
| 5,N                                                                                                                                              | 11,Z,O   | Signal control enable status (Enumeration/Logical) Z is the LCN enumeration position for the RTU 1-state. For example, to map control enable to MODE (MAN, AUTO), enter 11,1,0. (Control enable = true is the manual state, Control enable = false is the AUTO state. The LCN has Man = 0, Auto = 1.) |
| 8, size                                                                                                                                          | 13,R,C,N | Logical Array (R = row, C = column, N = number items) See note.                                                                                                                                                                                                                                       |
| 7, size                                                                                                                                          | 14,R,C,N | Analog Array (R = row, C = column, N = number items) See Note.                                                                                                                                                                                                                                        |
| 3                                                                                                                                                | 15       | Signal Questionable (signal state parameter).                                                                                                                                                                                                                                                         |
| <b>Note:</b> Enter array number in the AD1 field and leave the DPAR field blank. The maximum number of items in a single array conversion is 48. |          |                                                                                                                                                                                                                                                                                                       |

## 6.2.1 Check Before Operate (CBO)

---

### Introduction

The output signals of a digital composite image point effect the output signals of a Bristol Babcock digital board that is physically connected to the Bristol Babcock RTU. This board can be a digital output (DO) board or a Check Before Operate (CBO) DO board. The CBO functionality requires the CBO DO board.

---

### CBO DO configuration

The ACCOL load for the Bristol Babcock RTU must identify the type of board used. The ACCOL must also define the HCBO module for all of the CBO digital boards, and the CBO indicator for the DO must be set as **Y** through the Builder on the PITE. The HCBO module must be included in the RTU system task (TASK0). These DO signals can be latched (keep the output state until another output is given) or momentary (hold the output state for an identified time period then set it back to the normal state). The CLM CBO works with the ACCOL defined HCBO module only. Refer to the Bristol Babcock manual *ACCOL II Reference for PROM Set AD* or *ACCOL Software Revision 5.3* for detailed information on HCBO module configuration. The CBO function will apply to the RTU's CBO-type digital output boards only. DOs can be configured by the user with Toolkit to be latched or pulsed - that is, to have timed durations. For each DO point, the user configures the duration of the pulse. When an output is made, the output is maintained for the duration configured for that point; at the end of the duration, the output is automatically stopped. If the duration is configured to be 0, the output will be latched - i.e., it will not be automatically reset.

### Host-supervised CBO

The RTU's CBO DO boards will support two forms of CBO: host-supervised and local. Host-supervised CBO is the form in which multiple interactions between the CLM and the RTU are used to increase the probability that the outputs will be done properly. A brief and summarized description of host-supervised CBO is as follows. The CLM detects an operator or application request to make a digital output; and sends a SELECT request to the RTU. The RTU selects the outputs on the board, verifies that the SELECT occurred properly, and sends a SELECT response message to the CLM indicating success or the occurrence of an error. If the message from the RTU indicates success, the CLM sends an OPERATE message to the RTU. The RTU does the operate, checks for errors, and sends an OPERATE ACK to the CLM.

### Local CBO

With the local form of CBO, the RTU detects a request for a CBO output from an application within the RTU, from another RTU, or from the Toolkit, and performs all checks locally within the RTU without communicating to the requester to perform the checks. When the need to make an output is detected by the RTU, the RTU selects the outputs on the DO board, reads the intended outputs from the board, checks to make sure those outputs are consistent with both the selected and non-selected outputs, then commands the board to make the outputs.

---

*Continued on next page*

## 6.2.1 Check Before Operate (CBO), Continued

---

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CBO users and forms</b>     | <p>The CLM will use the host-supervised form of CBO for outputs requested by the TPS operator and by applications within the TPS system.</p> <p>The Bristol Babcock Toolkit and applications within RTUs in the network will use the local form of CBO. Local CBO will be used between the Toolkit and the RTU, even if communications are routed through the CLM.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>CBO between CLM and RTU</b> | <p>CBO between the CLM and RTU will work as follows:</p> <ul style="list-style-type: none"><li>• When a digital output is requested at the TPS - either by an operator or an application - the CLM sends a SELECT message to the RTU identifying the requested output. The RTU sends the output selection to the CBO DO circuit board.</li><li>• The RTU reads back from the CBO DO circuit board and verifies that it is prepared to make all outputs correctly - that is, the circuit board will make the output requested and will leave unchanged the outputs for which requests were not received. The RTU sends a SELECT response message to the CLM indicating that the DO board is prepared to make all outputs correctly.</li><li>• The CLM receives the SELECT response message from the RTU and verifies the following:<ul style="list-style-type: none"><li>a) The sequence number in the response message matches the sequence number in the SELECT message which it sent to the RTU.</li><li>b) The message came from the correct RTU.</li><li>c) The status from the RTU indicates that the SELECT activity at the RTU was successful (i.e., feedback from the board indicates all pending outputs are correct).</li></ul>The CLM sends to the RTU an OPERATE message, repeating the outputs identified in the original SELECT message.</li><li>• The RTU receives the OPERATE message, confirms that it identifies the same output identified in the original SELECT message and commands the CBO DO circuit board to make the outputs. The RTU verifies that the outputs actually made by the board are the same as those the board indicated it would make. If a discrepancy is found, the operation is terminated and the CLM is notified.</li></ul> <p>When an output is made for which a pulse duration is specified, the RTU will time the output and automatically reset the output at the end of the specified duration. The CLM is notified of the successful completion.</p> |

---

*Continued on next page*

## 6.2.1 Check Before Operate (CBO), Continued

---

### **CBO between CLM and RTU, continued**

The CLM, not the operator or the application that originated the CBO request, verifies the selection and issues the OPERATE command to the RTU. Once initiated by the operator or application, the activity will go to completion if no errors occur. Neither the operator or application will be able to approve or cancel the CBO activity after it has started.

### **CBO Timeout**

If the CBO activity times out within the RTU - that is, if the RTU does not receive an OPERATE message from the CLM within the time limit specified during configuration of the module - it will cancel that CBO activity.

### **CBO cancellation by RTU**

The RTU cancels the CBO activity as follows:

- a) clearing the current CBO request from its processing queue
- b) setting the bits in the select register equal to the current states of the outputs
- c) reporting an error to the CLM if it receives an OPERATE message from the CLM
- d) taking no further action on the activity

### **CBO cancellation by CLM**

If the CLM times out the CBO activity, it cancels the current CBO activity as follows:

- a) discards the current request from memory and from its processing queue
- b) issues an error message - and discards any pending SELECT requests since, because of the timeout delay, they may no longer be appropriate.

The length of the CLM timeout will be user configurable with a range of three to 15 minutes and a default value at ten minutes.

### **Concurrent CBO activities**

A CLM can perform concurrent CBO activities:

- with multiple RTUs
- with multiple boards within an RTU
- with multiple points on one board - to the extent supported by the throughput capacity of the system.

---

*Continued on next page*

## 6.2.1 Check Before Operate (CBO), Continued

---

### CBO queuing

If the RTU receives a local CBO request for a point on a board while it is processing a CBO activity for that point on the board, the local request is queued. The RTU processes the next queued request when the current CBO activity completes.

If the RTU receives a host-supervised CBO request for a point on a board while it is processing a local CBO activity for that point on the board, the host-supervised request is placed at the head of the queue and will be processed next.

The RTU has the ability to queue one SELECT message from the CLM for host-supervised CBO for each point on the CBO board. The RTU does not have a specific limit to the number of local CBO requests it can queue because it uses available memory to store them within the RTU.

### CBO failure

After the SELECT is made at the board, if the RTU determines that any of the outputs on the board will not be made correctly, it will fail those outputs and the board and notify the CLM. The board will not make any more outputs until it is taken out of the failed state. The board can be taken out of the failed state by being physically removed and replaced.

The board can also be taken out of the failed state as follows: With Toolkit, the user sets the failed CBO board's CLEAR ERRORS (or similarly named) signal and clears the board's diagnostic error array. The board's diagnostic array must be cleared in order for the CLM to attempt to communicate with the board. When output to the board is initiated at the TPS, the RTU receives the SELECT message and clears the board failure before it attempts the SELECT activity.

---

*Continued on next page*

## 6.2.1 Check Before Operate (CBO), Continued

---

### **CBO Command Disagree**

If there is a problem during the CBO activity, the CLM will issue a message unique and specific to the problem to the real time journal and the system printer at the TPS. A Command Disagree at the TPS will indicate that the operation did not occur. The message will be specific to the problem, identifying the CLM, RTU, and I/O board involved in the CBO activity.

The following problems will be covered in this manner:

1. The CLM did not receive a valid ACK to a command from the intended RTU or detected an ACK from an incorrect RTU.
2. At the RTU, the RTU I/O board indicates that its selected output does not match what is requested and expected for all eight outputs based on the SELECT request message from the CLM.
3. The RTU determines that the outputs identified in the SELECT request do not match the outputs identified in the OPERATE command.
4. The contact feed back from RTU I/O board does not match what was requested and expected for all eight bits.
5. There was loss of power to the board or the board was removed after the outputs were selected on the board and before the outputs were made.

### **CBO board removal or power loss**

If the CBO board loses power or is removed during a CBO activity, the CBO activity proceeds as follows when power is returned or the board is replaced:

- a) If the outputs were not selected on the CBO DO board, the CBO activity resumes and processes normally, provided that a CBO or power timeout (Refer to sections 6 and 13) has not occurred;
- b) If the power loss/board removal occurred after the outputs had been selected and before they were operated and the SELECT does not result in all eight outputs of the board being set to OFF, an error is reported by the CLM. If the board was removed or the power loss was momentary (i.e., power was restored before the OPERATE command was received from the CLM), the error message will be printed and journaled indicating power failure.

If the power loss was in effect when the CLM issued its OPERATE command, the CLM will time out and issue a message indicating that the RTU did not respond to the OPERATE message.

---

*Continued on next page*

## 6.2.1 Check Before Operate (CBO), Continued

---

### **CBO board restoration**

The RTU will have configurable parameters for identifying how outputs by the board are to be determined when power to the RTU is returned or a board is replaced after being removed. If the length of time without power or with the board removed exceeds the configured duration or if the board is configured for no RESTORE, all outputs are set to OFF when the board is made operational. If the length of time without power or with the board removed is less than the configured duration and the board is configured for RESTORE, the outputs are restored to their states at the time immediately prior to the power loss or board removal.

The range of the configurable duration parameter is 0 to 32767 seconds (0 to 9.19 hours, and values outside of that range are treated as 0). The determination of what outputs are to be made after power recovery/board replacement is made solely by the RTU - the CLM is not involved. The function of enabling/disabling the RESTORE is done only from the RTU PC Toolkit or load builder interface. The CLM will not provide that function.

---

## 6.3 Image Point Interface Details

### Image point composition

Each image point definition includes the following items:

#### Image Point Components

| Part | Description                               |
|------|-------------------------------------------|
| 1    | The point configuration                   |
| 2    | Custom Data Segment parameters (optional) |
| 3    | The builder input form                    |
| 4    | Supplied PITE template file               |

### PITE template assumptions

The PITE makes the following assumptions:

- RTU Logical signals are processed by exception. This requires the alarm enable bit in the RTU to be set for alarm signals and the RBE bit in the RTU to be set for RBE signals.
- RTU Logical control outputs will be scanned for security.
- The TRACK signal (control from the manual panel) is not supported at the TPS.
- Signals that do not change in operation, such as pipe diameter, are read by synchronization only.
- Each signal is used in at most one PITE.

### WARNING

Data from the RTU is processed using an RTU address lookup table. Each ACCOL signal name you assign and reference is automatically replaced by its equivalent RTU address during synchronization. If you create two PITEs using the same RTU signal, only one of them will work as intended, because there is only one address lookup table entry for each RTU signal used. Therefore, do not enter the same signal name in more than one PITE. AM connections allow you to reuse the data in additional AM points if required.

*Continued on next page*

## 6.3.1 AGA3 Flow Point (AM Regulatory Point Type)

### Introduction

AGA3 is a method for measuring natural gas flows using orifice meters.

### Point configuration

AGA3 Flow Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry  | Notes |
|-----------|-----------------------------|--------|-------|
| NOPINPTS  | Number of Input Connections | 1      |       |
| PISRC1    | Source Point ID.Parameter 1 | *.FLOW |       |
| PIDSTN1   | Destination Parameter       | P1     |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE |       |

*Continued on next page*

### 6.3.1 AGA3 Flow Point (AM Regulatory Point Type), Continued

#### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

#### General Input/Output Connections and Custom Data Segments (AM88-380)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks         | 0        |       |
| NOPKG     | No. of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches          | 0        |       |
| NOGINPTS  | Number of General Inputs    | 0        |       |
| NOGOPTS   | Number of General Outputs   | 0        |       |
| PKGNAME1  | Source File 1 Name          | B_AGA3   |       |
| PKGNAME2  | Source File 2 Name          | CLM_DMD3 |       |

#### CDS parameters

##### AGA3 Flow Point CDS Parameters (B\_AGA3)

| Parameter | ACCOL Signal               | Description                                        |
|-----------|----------------------------|----------------------------------------------------|
| DIFFPRES  | DIFF PRESS ( $h_w$ )*      | The measured pressure drop across an orifice plate |
| STATPRES  | STAT PRESS *               | Static pressure of flowing gas                     |
| ADJ_PRES  | ADJ PRESS                  | Average barometric pressure, psi                   |
| ORIF_D    | ORIF DIAM                  | Orifice diameter, inches                           |
| PIPE_D    | PIPE DIAM                  | Inside diameter of pipe, inches                    |
| ORIF_K    | ORIF CONST ( $K = F_a F_m$ | Orifice constant (defaults to 1.0)                 |
| BASE_P    | $F_1$ )                    | Base or contract pressure, psi                     |
| BASE_T    | BASE PRESS ( $P_b$ )       | Base or contract temperature, degF                 |
| FLOW_T    | BASE TEMP ( $T_b$ )        | Flowing temperature of gas, degF                   |
| FPV_IN    | FLOW TEMP ( $T_f$ )*       | Supercompressibility factor of gas                 |
| POINT     | FPV IN                     | Pipe tap or flange tap construction code           |
| SG        | POINT                      | Specific gravity of flowing gas (defaults to 0.6)  |
| FLOW      | TRACK                      | (not supported)                                    |
|           | FLOW OUTPUT ( $Q_h$ )*     | Corrected gas flow rate, MSCFH                     |

\* If RBE is used, these signals should be configured as RBE signals in the ACCOL build.

*Continued on next page*

### 6.3.1 AGA3 Flow Point (AM Regulatory Point Type), Continued

#### Builder input form

AGA3 Builder Input Form (B\_AGA3)

| Prompt    | Entry  | Description                                             |
|-----------|--------|---------------------------------------------------------|
| ENTRY     | B_AGA3 | Template Name (.ZT)                                     |
| NAME      |        | TPS Point Name                                          |
| PIT_NAME  |        | NDT PIT Table Name                                      |
| DIFFPRES  |        | DIFF PRESS RTU Signal Name                              |
| STATPRES  |        | STAT PRESS RTU Signal Name                              |
| ADJ_PRES  |        | ADJ PRESS RTU Signal Name                               |
| ORIF_D    |        | ORIF DIAM RTU Signal Name                               |
| PIPE_D    |        | PIPE DIAM RTU Signal Name                               |
| ORIF_K    |        | ORIF CONST RTU Signal Name                              |
| BASE_P    |        | BASE PRESS RTU Signal Name                              |
| BASE_T    |        | BASE TEMP RTU Signal Name                               |
| FLOW_T    |        | FLOW TEMP RTU Signal Name                               |
| FPV_IN    |        | FPV IN RTU Signal Name                                  |
| SPOINT    |        | POINT RTU Signal Name                                   |
| SG        |        | SPEC GRAV RTU Signal Name                               |
| FLOW      |        | FLOW OUTPUT RTU Signal Name                             |
| IO_SLOT 1 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 2 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 3 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 4 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 5 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 6 |        | RTV Slot number where I/O Card with signal is installed |
| END       |        | End of current entry                                    |

## 6.3.2 AGA5 Flow Point (AM Regulatory Point Type)

### Introduction

AGA5 is a method for determining natural gas energy from volume when no calorimeter signal is available.

### Point configuration

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

#### AGA5 Flow Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

#### AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| NOPINPTS  | Number of Input Connections | 1        |       |
| PISRC1    | Source Point ID.Parameter 1 | *.OUTPUT |       |
| PIDSTN1   | Destination Parameter       | P1       |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE   |       |

*Continued on next page*

## 6.3.2 AGA5 Flow Point (AM Regulatory Point Type), Continued

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

General Input/Out Connections and Custom Data Segments (AM88-380)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks         | 0        |       |
| NOPKG     | No. of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches          | 0        |       |
| NOGINPTS  | Number of General Inputs    | 0        |       |
| NOGOPTS   | Number of General Outputs   | 0        |       |
| PKGNAME1  | Source File 1 Name          | B_AGA5   |       |
| PKGNAME2  | Source File 2 Name          | CLM_DMD3 |       |

### CDS parameters

AGA5 Flow Point CDS Parameters (B\_AGA5)

| Parameter            | ACCOL Signal           | Notes                                                  |
|----------------------|------------------------|--------------------------------------------------------|
| VOLUME               | VOLUME *               | Volume of gas, MSCF or flow rate, MSCFH from meter     |
| BASE_P               | BASE PRESS             | Base pressure of flowing gas, psi (defaults to 14.73)  |
| BASE_T               | BASE TEMP              | Base temperature of flowing gas, degF (defaults to 60) |
| FPV_IN               | FPV IN                 | Supercompressibility factor of gas                     |
| SG                   | SPEC GRAV              | Specific gravity of flowing gas                        |
| VOL_CO <sub>2</sub>  | VOL % CO <sub>2</sub>  | % CO <sub>2</sub> in gas                               |
| VOL_N <sub>2</sub>   | VOL % N <sub>2</sub>   | % N <sub>2</sub> in gas                                |
| VOL_O <sub>2</sub>   | VOL % O <sub>2</sub>   | % O <sub>2</sub> in gas                                |
| VOL_He               | VOL % He               | % He in gas                                            |
| VOL_CO               | VOL % CO               | % CO in gas                                            |
| VOL_H <sub>2</sub> S | VOL % H <sub>2</sub> S | % H <sub>2</sub> S in gas                              |
| VOL_H <sub>2</sub> O | VOL % H <sub>2</sub> O | % H <sub>2</sub> O in gas                              |
| VOL_H <sub>2</sub>   | VOL % H <sub>2</sub>   | % H <sub>2</sub> in gas                                |
| VOL_CONV             | VOL CONVERS            | Volume conversion factor                               |
| ENRG_CNV             | ENERGY CONV            | Energy conversion factor                               |
| OUTPUT               | OUTPUT                 | Gas energy (module output)                             |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

*Continued on next page*

## 6.3.2 AGA5 Flow Point (AM Regulatory Point Type), Continued

### Builder input form

AGA5 Builder Input Form (B\_AGA5)

| Prompt               | Entry  | Description                                             |
|----------------------|--------|---------------------------------------------------------|
| ENTRY                | B_AGA5 | Template Name (.ZT)                                     |
| NAME                 |        | TPS Point Name                                          |
| PIT_NAME             |        | NDT PIT Table Name                                      |
| VOLUME               |        | VOLUME RTU Signal Name                                  |
| BASE_P               |        | BASE PRESS RTU Signal Name                              |
| BASE_T               |        | BASE TEMP RTU Signal Name                               |
| FPV_IN               |        | FPV IN RTU Signal Name                                  |
| SG                   |        | SPEC GRAV RTU Signal Name                               |
| VOL_CO <sub>2</sub>  |        | VOL % CO <sub>2</sub> RTU Signal Name                   |
| VOL_N <sub>2</sub>   |        | VOL % N <sub>2</sub> RTU Signal Name                    |
| VOL_O <sub>2</sub>   |        | VOL % O <sub>2</sub> RTU Signal Name                    |
| VOL_He               |        | VOL % He RTU Signal Name                                |
| VOL_CO               |        | VOL % CO RTU Signal Name                                |
| VOL_H <sub>2</sub> S |        | VOL % H <sub>2</sub> S RTU Signal Name                  |
| VOL_H <sub>2</sub> O |        | VOL % H <sub>2</sub> O RTU Signal Name                  |
| VOL_H <sub>2</sub>   |        | VOL % H <sub>2</sub> RTU Signal Name                    |
| VOL_CONV             |        | VOL CONVERTS RTU Signal Name                            |
| ENRG_CNV             |        | ENERGY CONV RTU Signal Name                             |
| OUTPUT               |        | OUTPUT RTU Signal Name                                  |
| IO_SLOT 1            |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 2            |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 3            |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 4            |        | RTV Slot number where I/O Card with signal is installed |
| END                  |        | End of current entry                                    |

### 6.3.3 AGA7 Flow Point (AM Regulatory Point Type)

#### Introduction

AGA7 is a method for measuring natural gas flow using a turbine flow meter, positive displacement meter, or vortex shedding meter.

#### Point configuration

AGA7 Flow Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

#### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| NOPINPTS  | Number of Input Connections | 1        |       |
| PISRC1    | Source Point ID.Parameter 1 | *.OUTPUT |       |
| PIDSTN1   | Destination Parameter       | P1       |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE   |       |

*Continued on next page*

### 6.3.3 AGA7 Flow Point (AM Regulatory Point Type), Continued

#### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution (AM88-370)."

General Input/Out Connections and Custom Data Segments (AM88-380)

| Parameter | Description               | Entry    | Notes |
|-----------|---------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks       | 0        |       |
| NOPKG     | Number of Custom Data     | 2        |       |
| NUMSWTCH  | Packages                  | 0        |       |
| NOGINPTS  | Number of Switches        | 0        |       |
| NOGOPTS   | Number of General Inputs  | 0        |       |
| PKGNAME1  | Number of General Outputs | B_AGA7   |       |
| PKGNAME2  | Source File 1 Name        | CLM_DMD3 |       |
|           | Source File 2 Name        |          |       |

#### CDS parameters

AGA7 Flow Point CDS Parameters (B\_AGA7)

| Parameter | ACCOL Signal  | Notes                                                                  |
|-----------|---------------|------------------------------------------------------------------------|
| FLOW_SW   | FLOW SWITCH * | Logical, select equation                                               |
| DENS_SW   | DENS SWITCH   | Logical, select densitometer or gravitometer to determine base density |
| FLOW_T    | FLOW TEMP *   | Flowing gas temperature, deg F (defaults to 60)                        |
| FLOW_P    | FLOW PRESS *  | Flowing gas pressure, psig                                             |
| RATE      | RATE *        | Uncorrected gas flow                                                   |
| BASE_T    | BASE TEMP     | Contract base temperature, deg F                                       |
| BASE_P    | BASE PRESS    | Contract base pressure, psia                                           |
| FPV_IN    | FPV IN        | Supercompressibility (defaults to 1.0)                                 |
| PA        | ADJ PRESS     | Barometric pressure, psia (defaults to 14.73)                          |
| FLOW_D    | FLOW DENS     | Density of flowing gas as measured by densitometer                     |
| BASE_D    | BASE DENS     | Contract base density                                                  |
| SG        | SPEC GRAV     | Specific gravity                                                       |
| GRAV_T    | GRAV TEMP     | Temperature at gravitometer, deg F                                     |
| GRAV_P    | GRAV PRESS    | Pressure at gravitometer, psia                                         |
| C         | CALIB FACTR   | Calibration factor (defaults to 1.0)                                   |
| K         | SPAN          | Span, adjust output to desired units                                   |
| OUTPUT    | OUTPUT *      | Calculated flow rate at base conditions                                |

\* If RBE is used, these signals should be configured as RBE signals in the ACCOL build.

*Continued on next page*

### 6.3.3 AGA7 Flow Point (AM Regulatory Point Type), Continued

#### Builder input form

AGA7 Builder Input Form (B\_AGA7)

| Prompt    | Entry  | Description                                             |
|-----------|--------|---------------------------------------------------------|
| ENTRY     | B_AGA7 | Template Name (.ZT)                                     |
| NAME      |        | TPS Point Name                                          |
| PIT_NAME  |        | NDT PIT Name                                            |
| FLOW_SW   |        | FLOW SWITCH RTU Signal Name                             |
| DENS_SW   |        | DENS SWITCH RTU Signal Name                             |
| FLOW_T    |        | FLOW TEMP RTU Signal Name                               |
| FLOW_P    |        | FLOW PRESS RTU Signal Name                              |
| RATE      |        | RATE RTU Signal Name                                    |
| BASE_T    |        | BASE TEMP RTU Signal Name                               |
| BASE_P    |        | BASE PRESS RTU Signal Name                              |
| FPV_IN    |        | FPV IN RTU Signal Name                                  |
| PA        |        | ADJ PRESS RTU Signal Name                               |
| FLOW_D    |        | FLOW DENS RTU Signal Name                               |
| BASE_D    |        | BASE DENS RTU Signal Name                               |
| SG        |        | SPEC GRAV RTU Signal Name                               |
| GRAV_T    |        | GRAV TEMP RTU Signal Name                               |
| GRAV_P    |        | GRAV PRESS RTU Signal Name                              |
| C         |        | CALIB FACTR RTU Signal Name                             |
| K         |        | SPAN RTU Signal Name                                    |
| OUTPUT    |        | OUTPUT RTU Signal Name                                  |
| IO_SLOT 1 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 2 |        |                                                         |
| IO_SLOT 3 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 4 |        |                                                         |
| IO_SLOT 5 |        | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 6 |        |                                                         |
| END       |        | End of current entry                                    |

## 6.3.4 ANIN, High- and Low-Level Analog Input (AM Regulatory Point Type)

### Introduction

ANIN

### Point configuration

Analog Input Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry   | Notes |
|-----------|-----------------------------|---------|-------|
| NOPINPTS  | Number of Input Connections | 1       |       |
| PISRC1    | Source Point ID.Parameter 1 | *.INPUT |       |
| PIDSTN1   | Destination Parameter       | P1      |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE  |       |

*Continued on next page*

## 6.3.4 ANIN,High- and Low-Level Analog Input (AM Regulatory Point Type), Continued

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution (AM88-370)."

General Input/Out Connections and Custom Data Segments (AM88-380)

| Parameter | Description                    | Entry    | Notes |
|-----------|--------------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks            | 0        |       |
| NOPKG     | Number of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches             | 0        |       |
| NOGINPTS  | Number of General Inputs       | 0        |       |
| NOGOPTS   | Number of General Outputs      | 0        |       |
| PKGNAME1  | Source File 1 Name             | B_ANIN   |       |
| PKGNAME2  | Source File 2 Name             | CLM_DMD3 |       |

### CDS parameters

Analog Input Point CDS Parameters (B\_ANIN)

| Parameter | ACCOL Signal | Notes                                             |
|-----------|--------------|---------------------------------------------------|
| INPUT     | INPUT *      |                                                   |
| ZERO      | ZERO         |                                                   |
| SPAN      | SPAN         |                                                   |
| DEVICE    |              | I/O Card Slot No. (Not supported by BSAP)         |
| INITIAL   |              | Field Wiring Terminal No. (Not supported by BSAP) |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

### Builder input form

Analog Input Builder Input Form (B\_ANIN)

| Prompt   | Entry  | Description                                 |
|----------|--------|---------------------------------------------|
| ENTRY    | B_ANIN | Template Name (.ZT)                         |
| NAME     |        | TPS Point Name                              |
| PIT_NAME |        | NDT PIT Name                                |
| INPUT    |        | INPUT RTU Signal Name                       |
| ZERO     |        | ZERO RTU Signal Name                        |
| SPAN     |        | SPAN RTU Signal Name                        |
| IO_SLOT1 |        | RTU Slot number where I/O card is installed |
| END      |        | End of current entry                        |

## 6.3.5 ANOUT, Analog Output (AM Regulatory Point Type)

### Introduction

ANOUT

### Point configuration

#### Analog Output Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes                   |
|-----------|-------------------------|----------|-------------------------|
| NAME      | Point Name              |          |                         |
| UNIT      | Unit ID                 |          |                         |
| PTDESC    | Point Descriptor        |          |                         |
| KEYWORD   | Point Keyword           |          |                         |
| EUDESC    | E.U. Descriptor         |          |                         |
| PRIMMOD   | Primary Module Point ID | ----     |                         |
| PTDISCL   | Point Disclosure        | FULL     |                         |
| PVALGID   | PV Algorithm            | NULL     |                         |
| CTLALGID  | Control Algorithm       | CL       |                         |
| OVERVAL   | Overview Display Value  |          |                         |
| SUPPIO    | I/O Display Suppression | INPSUPPR |                         |
| PERIOD    | Point Execution Period  | ----     | Point must be scheduled |

### Control mode configuration

#### AM Regulatory Control Mode and Setpoint Form (AM88-341)

| Prompt   | Description               | Entry             |
|----------|---------------------------|-------------------|
| NMODE    | Point's Normal Mode       | MAN               |
| NMODATTR | Normal Mode Attribute     | PROGRAM           |
| MODEPERM | Mode Permissive           | PERMIT            |
| EXTSWOPT | Ext Mode Switching Option | NONE              |
| SPFORMAT | SP Decimal Point Format   |                   |
| SPEUHI   | SP EU High Range In EUs   |                   |
| SPEULO   | SP EU Low Range In EUs    |                   |
| SPHILM   | SP High Limit             |                   |
| SPLOLM   | SP Low Limit              |                   |
| SP       | Set Point                 |                   |
| SPOPT    | Set Point Option          | None<br>TV<br>ASP |

*Continued on next page*

## 6.3.5 ANOUT, Analog Output (AM Regulatory Point Type),

Continued

### Control algorithm

AM Control Auto/Manual Algorithm (AM88-358)

| Parameter | Description                 | Entry               | Notes                      |
|-----------|-----------------------------|---------------------|----------------------------|
| CVTYPE    | Type of CV                  | Engrunit<br>Percent | (Default)                  |
| INITTYPE  | Initialization Type         | NONE                |                            |
| ARWDI     | Anti-Reset Windup Direction | DIRECT<br>REVERSE   |                            |
| NOCOPTS   | No. Output Connections      | 0                   |                            |
| CVEUHI    | CV Engineering Units High   | 0                   |                            |
| CVEULO    | CV Engineering Units Low    |                     |                            |
| OP        | Control Output Value        |                     |                            |
| OPHILM    | OP High Limit Value in %    |                     |                            |
| OPLOLM    | OP Low Limit Value in %     |                     |                            |
| OPMCHLM   | OP Minimum Charge in %      |                     |                            |
| OPROCLM   | OP Roc Limit In % / Min     | -----               | Point must be<br>scheduled |

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

General Input/Out Connections and Custom Data Segments (AM88-380)

| Parameter | Description                    | Entry    | Notes |
|-----------|--------------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks            | 2        |       |
| NOPKG     | Number of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches             | 0        |       |
| NOGINPTS  | Number of General Inputs       | 0        |       |
| NOGOPTS   | Number of General Outputs      | 0        |       |
| PKGNAME1  | Source File 1 Name             | B_ANOUT  |       |
| PKGNAME2  | Source File 2 Name             | CLM_DMD3 |       |

*Continued on next page*

## 6.3.5 ANOUT, Analog Output (AM Regulatory Point Type),

Continued

### CDS parameters

#### Analog Output Point CDS Parameters (B\_ANOUT)

| Parameters                                            | ACCOL Signal                    | Notes                                                                                                                                                                    |
|-------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OUTPUT<br>OP<br>ZERO<br>SPAN<br>DEVICE<br><br>INITIAL | RESET<br>OUTPUT<br>ZERO<br>SPAN | Read back of actual RTU output<br>Desired Output<br><br><br><br>Slot number of I/O board (Not supported by BSAP)<br>Field wiring terminal number (Not supported by BSAP) |

\* If RBE is used, signals should be configured as RBE signals in the ACCOL build.

### Builder input form

#### Analog Output Builder Input Form (B\_ANOUT)

| Prompt                                                                          | Entry   | Description                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENTRY<br>NAME<br>PIT_NAME<br>OUTPUT<br>ZERO<br>SPAN<br>RESET<br>IO_SLOT1<br>END | B_ANOUT | Template Name (.ZT)<br>TPS Point Name<br>NDT PIT Name<br>OUTPUT signal name<br>ZERO signal name<br>SPAN signal name<br>RESET signal name<br>RTU slot number where I/O card with signal is installed<br>End of current entry |

#### ATTENTION

The analog output image point requires that you link a CL block named ANOT\_ALG. This block moves the output signal as read back from the RTU to the SP parameter.

This CL block is included in the package B\_ANOUT.AO. An unlink/link file is installed in your CL directory. To link or relink this block, the form of the command is:

```
EC NET>UCLM>B_ANOTCL.EC PAR1
```

where **UCLM** is the CL directory named in the installation process, and **PAR1** is the name of the point. An error may result if the modules are not already linked to the point.

## 6.3.6 ARRAYI, ARRAYO, Array Input/Output

---

### Introduction

The CLM for Bristol Babcock RTUs supports both logical and analog arrays. The output of logical arrays is not supported in this release. A single PITE parameter can transfer up to 48 elements between a CDS array and an array in the RTU.

Since array points are application specific and unrelated to any normal data acquisition points, the construction and naming of CDS parameters is left for the user.

The array templates provided have three user entered elements per PITE parameter: The CDS array name, the RTU array number, and the conversion code. The conversion code includes the definition of row, column, and number of items. For arrays, each PITE parameter is limited to 48 elements.

---

### RTU rows, columns and large arrays

If the array on the RTU is a single dimensioned array, row number is the starting element number. In a two dimensioned array, all of the elements in one row are read, column by column, before the next row is started. For this to work correctly, the TPS parameter name must be the array name. Never use an element name to represent the array name.

EXAMPLE 1. Read array number 10, a 96 element single dimensioned array into two TPS array parameters, ARR1 and ARR2. Make a custom TDF file with two PITE parameters:

```
ARRAY1=ARR1
ARRAY_N1=10
P01_ICNV=7,48-14,1,1,48
ARRAY2=ARR2
ARRAY_N2=10
P02_ICNV=7,48-14,49,1,48
```

EXAMPLE 2. Read array number 11, a 2 row, 48 column two dimensioned array into two TPS array parameters, ARR1 and ARR2. Make a custom TDF file with two PITE parameters:

```
ARRAY1=ARR1
ARRAY_N1=11
P01_ICNV=7,48-14,1,1,48
ARRAY2=ARR2
ARRAY_N2=11
P01_ICNV=7,48-14,2,1,48
```

---

*Continued on next page*

## 6.3.6 ARRAYI, ARRAYO, Array Input/Output, Continued

**Point configuration** Point configuration for array points is left to the user. Arrays in any user written CDS may be used.

**CDS parameters** The user may assign any convenient name to arrays which will be named in an array PITE.

### Builder Input form

#### Analog Array Input Point Builder Input Form (B\_ARRAYI)

| Prompt   | Entry    | Description                                                                                                                                                           |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENTRY    | B_ARRAYI | Template Name (.ZT)                                                                                                                                                   |
| NAME     |          | TPS Point Name                                                                                                                                                        |
| PIT_NAME |          | NDT PIT Name                                                                                                                                                          |
| ARRAY    |          | User defined array existing in a CDS attached to the named point                                                                                                      |
| ARRAY_NO |          | The number of the array in the RTU                                                                                                                                    |
| PO1_ICNV |          | Input conversion string "7,N-14,R,C,N" where:<br>N is the number of elements in the array (48 max)<br>R is the starting row number<br>C is the starting column number |
| END      |          | End of current entry                                                                                                                                                  |

### Builder Input form

#### Analog Array Output Point Builder Input Form (B\_ARRAYO)

| Prompt   | Entry    | Description                                                                                                                                                            |
|----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENTRY    | B_ARRAYO | Template Name (.ZT)                                                                                                                                                    |
| NAME     |          | TPS Point Name                                                                                                                                                         |
| PIT_NAME |          | NDT PIT Name                                                                                                                                                           |
| ARRAY    |          | User defined array existing in a CDS attached to the named point                                                                                                       |
| ARRAY_NO |          | The number of the array in the RTU                                                                                                                                     |
| PO1_OCNV |          | Output conversion string "7,N-14,R,C,N" where:<br>N is the number of elements in the array (48 max)<br>R is the starting row number<br>C is the starting column number |
| END      |          | End of current entry                                                                                                                                                   |

## 6.3.7 HSCount/LSCount, Counter (Regulatory Point Type)

### Introduction

This point has two input values: count and frequency. You may select either one for PV display.

### Point configuration

Counter Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry             | Notes |
|-----------|-----------------------------|-------------------|-------|
| NOPINPTS  | Number of Input Connections | 1                 |       |
| PISRC1    | Source Point ID.Parameter 1 | *.FREQ<br>*.COUNT |       |
| PIDSTN1   | Destination Parameter       | P1                |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE            |       |

*Continued on next page*

### 6.3.7 HSCount/LSCount (AM Regulatory Point Type), Continued

#### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution (AM88-370)."

General Input/Output Connections and CDSs (AM88-380)

| Parameter | Description                    | Entry    | Notes |
|-----------|--------------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks            | 0        |       |
| NOPKG     | Number of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches             | 0        |       |
| NOGINPTS  | Number of General Inputs       | 0        |       |
| NOGOPTS   | Number of General Outputs      | 0        |       |
| PKGNAME1  | Source File 1 Name             | B_COUNT  |       |
| PKGNAME2  | Source File 2 Name             | CLM_DMD3 |       |

#### CDS parameters

Counter Point CDS Parameters (B\_COUNT)

| Parameters | ACCOL Signal | Notes                                                   |
|------------|--------------|---------------------------------------------------------|
| COUNT      | COUNT *      | accumulated count                                       |
| C_ZERO     | COUNT ZERO   | scale count to 0% (0 default)                           |
| C_SPAN     | COUNT SPAN   | multiplier                                              |
| RESET      | RESET *      | resets the counter                                      |
| FREQ       | FREQUENCY    | frequency of input signal                               |
| F_ZERO     | FREQ ZERO    | 0% value of frequency                                   |
| F_SPAN     | FREQ SPAN    | frequency multiplier                                    |
| DEVICE     |              | Slot number of remote I/O board (Not supported by BSAP) |
| INITIAL    |              | Field wiring terminal number (Not supported by BSAP)    |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

*Continued on next page*

## 6.3.7 HSCount/LSCount (AM Regulatory Point Type), Continued

### Builder input form

Counter Point Builder Input Form (B\_COUNT)

| Prompt   | Entry   | Description                                 |
|----------|---------|---------------------------------------------|
| ENTRY    | B_COUNT | Template Name (.ZT)                         |
| NAME     |         | TPS Point Name                              |
| PIT_NAME |         | NDT PIT Name                                |
| COUNT    |         | COUNT RTU Signal Name                       |
| C_ZERO   |         | COUNT ZERO RTU Signal Name                  |
| C_SPAN   |         | COUNT SPAN RTU Signal Name                  |
| RESET    |         | RESET RTU Signal Name                       |
| FREQ     |         | FREQUENCY RTU Signal Name                   |
| F_ZERO   |         | FREQ ZERO RTU Signal Name                   |
| F_SPAN   |         | FREQ SPAN RTU Signal Name                   |
| IO_SLOT1 |         | RTU slot number where I/O card is installed |
| END      |         | End of current entry                        |

## 6.3.8 Digital Composite Points

### Introduction

The CLM Digital Composite point type is a CLM Platform based function.

### Point configuration

Please reference the *CLM Platform Implementation Manual* for information on building a CLM Digital Composite point. This section only contains information for mapping the data for an application specific device.

### Alarming

Digital composite point runtime errors (configuration and operation) are listed in the *CLM Platform Implementation Manual*.

#### ATTENTION

Refer to the Digital Composite Image Point section starting at 8.2.2 in the *CLM Platform Implementation Manual* for more information on the implementation of the digital composite point including CL modules. Also refer to 6.2.1 for CBO information.

## 6.3.8.1 Digital Composite Points (3-State) (AM Switch Point Type)

### Introduction

The 3-state digital composite point assembles up to one ACCOL input signals and up to two ACCOL output signals. It uses the generic digital composite point described in the *CLM Platform Implementation Manual*.

### Point configuration

Include CDS package **B\_DG3CDS** for this specific application.

### 3-State Digital Point

Digital Composite Point CDS parameters need to be considered and defined either by using the default value or modifying the value as needed on this point. The CDS B\_DG3CDS must be attached to the 3-state digital point in addition to DIG3CDS, DIG3CDS1, and CLM\_DMD3 (NOPKG=4).

BSAP User-Entered CDS Constants - B\_DG3CDS

| Parameter | Notes                               |
|-----------|-------------------------------------|
| DEVICEI1  | RTU Slot Number, Input 1            |
| INIT1     | Field Wiring Terminal No., Input 1  |
| DEVICEO1  | RTU Slot Number, Output 1           |
| INITO1    | Field Wiring Terminal No., Output 1 |
| DEVICEO2  | RTU Slot Number, Output 2           |
| INITO2    | Field Wiring Terminal No., Output 2 |

### Builder input form

3-State Digital Composite Builder Input Form (B\_DIG3)

| Prompt   | Entry  | Description                                 |
|----------|--------|---------------------------------------------|
| ENTRY    | B_DIG3 | Template Name (.ZT)                         |
| NAME     |        | TPS Point Name                              |
| PIT_NAME |        | NDT PIT Name                                |
| D1       |        | INPUT, input 1 RTU Signal Name              |
| OP1      |        | OUTPUT, output 1 RTU Signal Name            |
| CBO      |        | Check Before Operate indicator (Y or N)     |
| RESET1   |        | RESET, reset 1 RTU Signal Name              |
| OP2      |        | OUTPUT, output 2 RTU Signal Name            |
| CBO      |        | Check Before Operate indicator (Y or N)     |
| RESET2   |        | RESET, reset 2 RTU Signal Name              |
| IO_SLOT1 |        | RTU slot number where I/O card is installed |
| IO_SLOT2 |        | RTU slot number where I/O card is installed |
| IO_SLOT3 |        | RTU slot number where I/O card is installed |
| END      |        | End of current entry                        |

*Continued on next page*

## 6.3.8.2 Digital Composite Point (4-State) (AM Switch Point Type)

---

### Introduction

The 4-state digital composite point assembles up to two ACCOL input signals and up to three ACCOL output signals. It uses the generic digital composite point described in the *CLM Platform Implementation Manual*.

---

### Point configuration

Include CDS package **B\_DG4CDS** for this specific application.

---

*Continued on next page*

## 6.3.8.2 Digital Composite Point (4-State) (AM Switch Point Type), Continued

### 4-State Digital Point (DIG4CDS)

Digital Composite Point CDS Parameters need to be considered and defined either by using the default value or modifying the value as needed on this point. The CDS B\_DG4CDS must be attached to the 4-state digital point in addition to DIG4CDS, DIG4CDS1, and CLM\_DMD3 (NOPKG=4).

BSAP User-Entered CDS Constants - B\_DG4CDS

| Parameter | Notes                               |
|-----------|-------------------------------------|
| DEVICEI1  | RTU Slot Number, Input 1            |
| INIT1     | Field Wiring Terminal No., Input 1  |
| DEVICEI2  | RTU Slot Number, Input 2            |
| INIT2     | Field Wiring Terminal No., Input 2  |
| DEVICEO1  | RTU Slot Number, Output 1           |
| INITO1    | Field Wiring Terminal No., Output 1 |
| DEVICEO2  | RTU Slot Number, Output 2           |
| INITO2    | Field Wiring Terminal No., Output 2 |
| DEVICEO3  | RTU Slot Number, Output 3           |
| INITO3    | Field Wiring Terminal No., Output 3 |

### Builder input form

#### 4-State Digital Composite Builder Input Form (B\_DIG4)

| Prompt   | Entry  | Description                                 |
|----------|--------|---------------------------------------------|
| ENTRY    | B_DIG4 | Template Name (.ZT)                         |
| NAME     |        | TPS Point Name                              |
| PIT_NAM  |        | NDT PIT Name                                |
| E        |        | INPUT, input 1 RTU Signal Name              |
| D1       |        | INPUT, input 2 RTU Signal Name              |
| D2       |        | OUTPUT, output 1 RTU Signal Name            |
| OP1      |        | Check Before Operate indicator (Y or N)     |
| CBO      |        | RESET, output 1 RTU Signal Name             |
| RESET1   |        | OUTPUT, output 2 RTU Signal Name            |
| OP2      |        | Check Before Operate indicator (Y or N)     |
| CBO      |        | RESET, output 2 RTU Signal Name             |
| RESET2   |        | OUTPUT, output 3 RTU Signal Name            |
| OP3      |        | Check Before Operate indicator (Y or N)     |
| CBO      |        | RESET, output 3 RTU Signal Name             |
| RESET3   |        | RTU slot number where I/O card is installed |
| IO_SLOT1 |        | RTU slot number where I/O card is installed |
| IO_SLOT2 |        | RTU slot number where I/O card is installed |
| IO_SLOT3 |        | RTU slot number where I/O card is installed |
| IO_SLOT4 |        | RTU slot number where I/O card is installed |
| IO_SLOT5 |        | End of current entry                        |
| END      |        |                                             |

### 6.3.8.3 Digital Composite Point (5-State) (AM Switch Point Type)

---

#### Introduction

The 5-state digital composite point assembles up to two ACCOL input signals and up to three ACCOL output signals. It uses the generic digital composite point described in the *CLM Platform Implementation Manual*.

---

#### Point configuration

Include CDS package **B\_DG5CDS** for this specific application.

---

*Continued on next page*

### 6.3.8.3 Digital Composite Point (5-State) (AM Switch Point Type), Continued

#### 5-State Digital Point (DIG5CDS)

Digital Composite Point CDS Parameters need to be considered and defined either by using the default value or modifying the value as needed on this point. The CDS B\_DG5CDS must be attached to the 5-state digital point in addition to DIG5CDS, DIG5CDS1, and CLM\_DMD3 (NOPKG=4)..

#### BSAP User-Entered CDS Constants - B\_DG5CDS

| Parameter | Notes                               |
|-----------|-------------------------------------|
| DEVICEI1  | RTU Slot Number, Input 1            |
| INIT1     | Field Wiring Terminal No., Input 1  |
| DEVICEI2  | RTU Slot Number, Input 2            |
| INIT2     | Field Wiring Terminal No., Input 2  |
| DEVICEO1  | RTU Slot Number, Output 1           |
| INITO1    | Field Wiring Terminal No., Output 1 |
| DEVICEO2  | RTU Slot Number, Output 2           |
| INITO2    | Field Wiring Terminal No., Output 2 |
| DEVICEO3  | RTU Slot Number, Output 3           |
| INITO3    | Field Wiring Terminal No., Output 3 |

#### Builder input form

#### 5-State Digital Composite Builder Input Form (B\_DIG5)

| Prompt   | Entry  | Description                                             |
|----------|--------|---------------------------------------------------------|
| ENTRY    | B_DIG5 | Template Name (.ZT)                                     |
| NAME     |        | TPS Point Name                                          |
| PIT_NAM  |        | NDT PIT Name                                            |
| E        |        | INPUT, input 1 RTU Signal Name                          |
| D1       |        | INPUT, input 2 RTU Signal Name                          |
| D2       |        | OUTPUT, output 1 RTU Signal Name                        |
| OP1      |        | Check Before Operate Indicator (Y or N)                 |
| CBO      |        | RESET, output 1 RTU Signal Name                         |
| RESET1   |        | OUTPUT, output 2 RTU Signal Name                        |
| OP2      |        | Check Before Operate Indicator (Y or N)                 |
| CBO      |        | RESET, output 2 RTU Signal Name                         |
| RESET2   |        | OUTPUT, output 3 RTU Signal Name                        |
| OP3      |        | Check Before Operate Indicator (Y or N)                 |
| CBO      |        | RESET, output 3 RTU Signal Name                         |
| RESET3   |        | RTU slot number where I/O card with signal is installed |
| IO_SLOT1 |        | RTU slot number where I/O card with signal is installed |
| IO_SLOT2 |        | RTU slot number where I/O card with signal is installed |
| IO_SLOT3 |        | RTU slot number where I/O card with signal is installed |
| IO_SLOT4 |        | RTU slot number where I/O card with signal is installed |
| IO_SLOT5 |        | End of current entry                                    |
| END      |        |                                                         |

*Continued on next page*

## 6.3.9 Digital Input Points

---

**Introduction** The CLM Digital Input point type is a CLM Platform based function.

---

**Point configuration** Please reference the *CLM Platform Implementation Manual* for information on building a CLM Digital Input point. This section only contains information for mapping the data for an application specific device.

---

### 6.3.9.1 DIN3, Digital Input 3-State Builder Input

#### Builder input form

Digital Input Point Builder Input Form (B\_DIN3)

| Prompt   | Entry  | Description                                             |
|----------|--------|---------------------------------------------------------|
| ENTRY    | B_DIN3 | Template Name (.ZT)                                     |
| NAME     |        | TPS Point Name                                          |
| PIT_NAME |        | NDT PIT Name                                            |
| INPUT    |        | INPUT RTU Signal Name                                   |
| IO_SLOT1 |        | RTU slot number where I/O card with signal is installed |
| END      |        | End of current entry                                    |

### 6.3.9.2 DIN4, Digital Input 4-State Builder Input

#### Builder input form

Digital Input Point Builder Input Form (B\_DIN4)

| Prompt   | Entry  | Description                                             |
|----------|--------|---------------------------------------------------------|
| ENTRY    | B_DIN4 | Template Name (.ZT)                                     |
| NAME     |        | TPS Point Name                                          |
| PIT_NAME |        | NDT PIT Name                                            |
| INPUT1   |        | First INPUT RTU Signal Name                             |
| INPUT2   |        | Second INPUT RTU Signal Name                            |
| IO_SLOT1 |        | I/O slot number where I/O card with signal is installed |
| IO_SLOT2 |        | I/O slot number where I/O card with signal is installed |
| END      |        | End of current entry                                    |

*Continued on next page*

## 6.3.10 FLAG, Flag Point (AM Flag Point Type)

### Introduction

The alarm enable bit or RBE in the RTU must be set for the INPUT signal. CDS is not supported for this point type.

TPS operator-entered values will be transmitted to the RTU.

### Point configuration

#### Flag Point Configuration (AM88-301)

| Parameter | Description             | Entry        | Notes |
|-----------|-------------------------|--------------|-------|
| NAME      | Point Name              |              |       |
| UNIT      | Unit ID                 |              |       |
| PTDESC    | Point Descriptor        |              |       |
| KEYWORD   | Point Keyword           |              |       |
| PRIMMOD   | Primary Module Point ID | -----        |       |
| PVSTATES0 | Upper Box Text          |              |       |
| PVSTATES1 | Lower Box Text          |              |       |
| UBOXCLR   | Upper Box Color         | Users Choice |       |
| LBOXCLR   | Lower Box Color         | Users Choice |       |
| OFFNRMAL  | Off Normal Alarming     | OFF<br>ON    |       |

### Builder input form

#### Flag Point Builder Input Form (B\_FLAG)

| Prompt   | Entry  | Description          |
|----------|--------|----------------------|
| ENTRY    | B_FLAG | Template Name (.ZT)  |
| NAME     |        | TPS Point Name       |
| PIT_NAME |        | NDT PIT Name         |
| PV *     |        | Input connection     |
| END      |        | End of current entry |

\* If RBE is used, this ACCOL signal associated with the flag parameter value should be configured as an RBE signal in the ACCOL build.

## 6.3.11 FPV, Supercompressibility Point (AM Regulatory Point Type)

### Introduction

FPV

### Point configuration

FPV Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Description       |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALIGID  | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | OUTSUPPR |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| NOPINPTS  | Number of Input Connections | 1        |       |
| PISRC1    | Source Point ID.Parameter 1 | *.OUTPUT |       |
| PIDSTN1   | Destination Parameter       | P1       |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE   |       |

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

General Input/Output Connections and Custom Data Segments (AM88-380)

| Parameter | Description                    | Entry    | Notes |
|-----------|--------------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks            | 0        |       |
| NOPKG     | Number of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches             | 0        |       |
| NOGINPTS  | Number of General Inputs       | 0        |       |
| NOGOPTS   | Number of General Outputs      | 0        |       |
| PKGNAME1  | Source File 1 Name             | B_FPV    |       |
| PKGNAME2  | Source File 2 Name             | CLM_DMD3 |       |

*Continued on next page*

## 6.3.11 FPV, Supercompressibility Point (AM Regulatory Point Type), Continued

### CDS parameters

#### FPV Point CDS Parameters (B\_FPV)

| Parameter | ACCOL Signal | Notes                                |
|-----------|--------------|--------------------------------------|
| FLOW_T    | FLOW TEMP *  | Flowing temp. of gas, defg           |
| STATPRES  | STAT PRESS   | Static pres. of gas, psig.           |
| SG        | SPEC GRAV    | Specific gravity of gas              |
| CO2MOLE   | CO2 MOLE     | CO2 content mole in %                |
| NMOLE     | N MOLE       | Nitrogen content mole in %           |
| OUTPUT    | OUTPUT *     | Computed supercompressibility factor |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

### Builder input form

#### FPV Builder Input Form (B\_FPV)

| Prompt    | Entry | Description                                             |
|-----------|-------|---------------------------------------------------------|
| ENTRY     | B_FPV | Template Name (.ZT)                                     |
| NAME      |       | TPS Point Name                                          |
| PIT_NAM   |       | NDT PIT Name                                            |
| FLOW_T    |       | FLOW TEMP RTU Signal Name                               |
| STATPRE   |       | STAT PRESS RTU Signal Name                              |
| SG        |       | SPEC GRAV RTU Signal Name                               |
| CO2MOLE   |       | CO2 MOLE RTU Signal Name                                |
| NMOLE     |       | N MOLE RTU Signal Name                                  |
| OUTPUT    |       | OUTPUT RTU Signal Name                                  |
| IO_SLOT 1 |       | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 2 |       | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 3 |       | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 4 |       | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 5 |       | RTV Slot number where I/O Card with signal is installed |
| IO_SLOT 6 |       | RTV Slot number where I/O Card with signal is installed |
| END       |       | End of current entry                                    |

*Continued on next page*

## 6.3.12 HWSTI, Smart Transmitter Point (AM Regulatory Point Type)

### Introduction

HWSTI

### Point configuration

Smart Transmitter Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Descriptor        |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALGID   | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | NULL     |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | NOSUPPR  |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry   | Notes |
|-----------|-----------------------------|---------|-------|
| NOPINPTS  | Number of Input Connections | 1       |       |
| PISRC1    | Source Point ID.Parameter 1 | *.INPUT |       |
| PIDSTN1   | Destination Parameter       | P1      |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE  |       |

*Continued on next page*

## 6.3.12 HWSTI, Smart Transmitter Point (AM Regulatory Point Type), Continued

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

General Input/Output Connections and Custom Data Segments (AM88-380)

| Parameter | Description                    | Entry    | Notes |
|-----------|--------------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks            | 0        |       |
| NOPKG     | Number of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches             | 0        |       |
| NOGINPTS  | Number of General Inputs       | 0        |       |
| NOGOPTS   | Number of General Outputs      | 0        |       |
| PKGNAME1  | Source File 1 Name             | B_HWSTI  |       |
| PKGNAME2  | Source File 2 Name             | CLM_DMD3 |       |

### Build visible CDS parameters

Build Visible Smart Transmitter CDS Parameters (B\_HWSTI)

| Parameter | Description              | Entry | Notes                                                                                             |
|-----------|--------------------------|-------|---------------------------------------------------------------------------------------------------|
| SENSRTYP  | Sensor Type              | 11.0  | 8 diff pres (DP)<br>9 gauge pres (GP)<br>10 abs pres (AP)<br>11 temp (STT)<br>12 MagneW3000 (SFM) |
| DECONF    | DE Configuration Mode    | 4.00  | 1 PV Only (PV)<br>2 PV & SV (PV_SV)<br>3 PV & data (PV_DB)<br>4 PV, SV, database (PV_SV_DB)       |
| PVCHAR    | Xmtr PV Characterization |       | 0 - 23                                                                                            |

*Continued on next page*

## 6.3.12 HWSTI, Smart Transmitter Point (AM Regulatory Point Type), Continued

### Complete listing of CDS parameters

Smart Transmitter Point CDS Parameters (B\_HWSTI)

| Parameter     | ACCOL Signal | Description                                                                  |
|---------------|--------------|------------------------------------------------------------------------------|
| STITAG        | STITAG       | Transmitter tag name (8 char)                                                |
| PVCHAR        | PVCHAR       | PV Characterization Option                                                   |
| MISMATCH (5)  |              | PVCHAR questionable                                                          |
| DECONF        | DECONF       | Digitally Enhanced Configuration Mode                                        |
| MISMATCH (3)  |              | DECONF questionable                                                          |
| URV           | URV          | Upper range value                                                            |
| MISMATCH (11) |              | URV questionable                                                             |
| LRV           | LRV          | Lower range value                                                            |
| MISMATCH (10) |              | LRV questionable                                                             |
|               |              | <b>Note:</b> User may provide alternate mapping if range change is required. |
| DAMPING       | DAMPING      | First order PV filtering option                                              |
| MISMATCH (4)  |              | DAMPING questionable                                                         |
| SENSRTYP      | SENSRTYP     | Smart transmitter sensor type                                                |
| MISMATCH (2)  |              | SENSRTYP questionable                                                        |
| STI_EU        | STIEU        | Transmitter Engineering Units                                                |
| SCOMMAND      | COMMAND      | Database transfer command                                                    |
| FREQ6050      | FREQ50       | Local power line frequency (STT only)                                        |
| PIUOTDCF      | PIUOTDCF     | Open Thermocouple Detect Enable (STT only)                                   |
| CJTACT        | CJTACT       | Internal Cold Junction Temp. Enable (STT)                                    |
| SSTATE        | STATUS       | STI Processor                                                                |
| INPUT         | PV*          | PV in URV/LRV Engineering Units                                              |
| SECVAR        | SECVAR*      | Secondary variable                                                           |
| URL           | URL          | Upper Range Limit (set at factory)                                           |
| MISMATCH (12) |              | URL questionable                                                             |
| LRL           | LRL          | Lower Range Limit (set at factory)                                           |
| STISWVER      | STISWVER     | Smart Xmtr Software revision level (8 char)                                  |
| SERIALNO      | SERIALNO     | Serial Number (STT or Magnew) or                                             |
| SCRCHPD       | SCRATCHPAD   | PROM Number                                                                  |
| XMITSTAT      | XMITSTAT     | (pressure transmitter) - 8 char                                              |
|               |              | 32 Character String - - S1(1)                                                |
|               |              | 64 Character String (truncated to 40) - -                                    |
| CFGSTAT       | CFGSTAT      | S1(2), S1(3)                                                                 |
|               |              | 32 Character String - - S1(4)                                                |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

*Continued on next page*

## 6.3.12 HWSTI, Smart Transmitter Point (AM Regulatory Point Type), Continued

CDS parameters  
unique to Bristol  
Babcock

### CDS Parameters Unique to Bristol Babcock HWSTI

| Parameter   | ACCOL Signal | Notes                                                                |
|-------------|--------------|----------------------------------------------------------------------|
| DONE        | DONE         | Completion of commands 1 through 8                                   |
| MISMATCH(1) | MISMATCH*    | Any of 11 configuration words mismatch                               |
| COMERRS     | COMERRS      | Count of transmitter communication errors detected                   |
| PWRFAIL     | POWERFAIL    | Slot number in card cage for HWSTI module<br>(Not supported by BSAP) |
| DEVICE      |              |                                                                      |
| CHANNEL     |              |                                                                      |

\* If RBE is used, this signal should be configured as an RBE signal in the ACCOL build.

*Continued on next page*

## 6.3.12 HWSTI, Smart Transmitter Point (AM Regulatory Point Type), Continued

### Builder input form

HWSTI Builder Input Form (B\_HWSTI)

| Prompt   | Entry   | Description                                             |
|----------|---------|---------------------------------------------------------|
| ENTRY    | B_HWSTI | Template Name (.ZT)                                     |
| NAME     |         | TPS Point Name                                          |
| PIT_NAME |         | NDT PIT Name                                            |
| PV       |         | PV RTU Signal Name                                      |
| SECVAR   |         | SECVAR RTU Signal Name                                  |
| STITAG   |         | STITAG RTU Signal Name                                  |
| PVCHAR   |         | PVCHAR RTU Signal Name                                  |
| DECONF   |         | DECONF RTU Signal Name                                  |
| URV      |         | URV RTU Signal Name                                     |
| LRV      |         | LRV RTU Signal Name                                     |
| DAMPING  |         | DAMPING RTU Signal Name                                 |
| SENSRTYP |         | SENSRTYP RTU Signal Name                                |
| COMMAND  |         | COMMAND RTU Signal Name                                 |
| FREQ6050 |         | FREQ50 RTU Signal Name                                  |
| PIUOTDCF |         | PIUOTDCF RTU Signal Name                                |
| CJTACT   |         | CJTACT RTU Signal Name                                  |
| URL      |         | URL RTU Signal Name                                     |
| LRL      |         | LRL RTU Signal Name                                     |
| STISWVER |         | STISWVER RTU Signal Name                                |
| SERIALNO |         | SERIALNO RTU Signal Name                                |
| SCRCHPAD |         | SCRATCHPAD RTU Signal Name                              |
| DONE     |         | DONE RTU Signal Name                                    |
| STATUS   |         | STATUS RTU Signal Name                                  |
| STI_EU   |         | STI_EU RTU Signal Name                                  |
| MISMATCH |         | MISMATCH RTU Signal Name                                |
| XMITSTAT |         | XMITSTAT RTU Signal Name                                |
| CFGSTAT  |         | CFGSTAT RTU Signal Name                                 |
| COMERRS  |         | COMERRS RTU Signal Name                                 |
| PWRFAIL  |         | POWERFAIL RTU Signal Name                               |
| IO_SLOT1 |         | RTU slot number where I/O card with signal is installed |
| END      |         | End of current entry                                    |

## 6.3.13 Numeric Point (AM Numeric Point Type)

### Introduction

TPS operator-entered values will be transmitted to the RTU.  
CDS is not supported for the AM Numeric Point.

### Point configuration

#### AM Numeric Data Point Configuration (AM88-320A)

| Parameter | Description             | Entry | Notes |
|-----------|-------------------------|-------|-------|
| NAME      | Point Name              | ----- |       |
| UNIT      | Unit ID                 |       |       |
| PTDESC    | Point Descriptor        |       |       |
| KEYWORD   | Point Keyword           |       |       |
| PRIMMOD   | Primary Module Point ID |       |       |
| EUDESC    | E. U. Descriptor        |       |       |
| RANGEHI   | PV Range HI             |       |       |
| RANGELO   | PV Range LOW            |       |       |
| PVFORMAT  | PV Decimal Point Format |       |       |
| PV        | PV                      |       |       |

### Builder input form

#### Numeric Point Builder Input Form (B-NUMER)

| Prompt   | Entry   | Description          |
|----------|---------|----------------------|
| ENTRY    | B_NUMER | Template Name (.ZT)  |
| NAME     |         | TPS Point Name       |
| PIT_NAME |         | NDT PIT Name         |
| PV *     |         | Input connection     |
| END      |         | End of current entry |

\* If RBE is used, this ACCOL signal associated with the numeric parameter value should be configured as an RBE signal in the ACCOL build.

## 6.3.14 PID3TM, PID Controller Point (AM Regulatory Point Type)

### Introduction

PID Controller

### Point configuration

PID Controller Point Configuration (AM88-301)

| Parameter | Description             | Entry    | Notes |
|-----------|-------------------------|----------|-------|
| NAME      | Point Name              |          |       |
| UNIT      | Unit ID                 |          |       |
| PTDESC    | Point Descriptor        |          |       |
| KEYWORD   | Point Keyword           |          |       |
| EUDESC    | E.U. Descriptor         |          |       |
| PRIMMOD   | Primary Module Point ID | ----     |       |
| PTDISCL   | Point Disclosure        | FULL     |       |
| PVALGID   | PV Algorithm            | DATAACQ  |       |
| CTLALGID  | Control Algorithm       | CL       |       |
| OVERVAL   | Overview Display Value  |          |       |
| SUPPIO    | I/O Display Suppression | NOSUPPR  |       |
| PERIOD    | Point Execution Period  | NOPERIOD |       |

### Data acquisition algorithm

Refer to Form "Regulatory PV Algorithms Common Display, AM88-302."

AM Regulatory Data Acquisition Algorithm (AM88-311)

| Parameter | Description                 | Entry   | Notes |
|-----------|-----------------------------|---------|-------|
| NOPINPTS  | Number of Input Connections | 1       |       |
| PISRC1    | Source Point ID.Parameter 1 | *.INPUT |       |
| PIDSTN1   | Destination Parameter       | P1      |       |
| PIACTSTS1 | Dest Conn Activity Status 1 | ACTIVE  |       |

*Continued on next page*

## 6.3.14 PID3TM, PID Continued

### Control mode configuration

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

AM Regulatory Control Mode and Setpoint Form (AM88-341)

| Parameter | Description               | Entry          | Notes |
|-----------|---------------------------|----------------|-------|
| NMODE     | Point's Normal Mode       | AUTO           |       |
| NMODATTR  | Normal Mode Attribute     | None           |       |
|           |                           | Operator       |       |
|           |                           | Program Permit |       |
| MODEPERM  | Mode Permissive           | Notperm        |       |
|           |                           | None           |       |
| EXTSWOPT  | Ext Mode Switching Option | EMS            |       |
|           |                           | EMP            |       |
| SPFORMAT  | SP Decimal Point Format   |                |       |
| SPEUHI    | SP EU High Range In EUs   |                |       |
| SPEULO    | SP EU Low Range In EUs    |                |       |
| SPHILM    | SP High Limit             |                |       |
| SPLOLM    | SP Low Limit              |                |       |
| SP        | Set Point                 |                |       |
| SPOPT     | Set Point Option          | None           |       |
|           |                           | TV             |       |
|           |                           | ASP            |       |

*Continued on next page*

## 6.3.14 PID3TM, PID Controller, Continued

### Control mode configuration, continued

#### AM Control CL Algorithm (AM88-364)

| Parameter | Description                 | Entry               | Notes   |
|-----------|-----------------------------|---------------------|---------|
| CVTYPE    | Type of CV                  | Engrunit<br>Percent | Default |
| INITTYPE  | Initialization Type         | NONE                |         |
| ARWDI     | Anti-Reset Windup Direction | DIRECT<br>REVERSE   |         |
| NOCOPTS   | No. of Output Connections   | 0                   |         |
| CVEUHI    | CV Engineering Units High   | 0                   |         |
| CVEULO    | CV Engineering Units Low    |                     |         |
| OP        | Control Output              |                     |         |
| OPHILM    | OP High Limit Value in %    |                     |         |
| OPLOLM    | OP Low Limit Value in %     |                     |         |
| OPMCHLM   | OP Minimum Change in %      |                     |         |

### I/O links and CDSs

Refer to Form "Regulatory Alarming and Distribution, AM88-370."

#### General Input/Out Connections and Custom Data Segments (AM88-380)

| Parameter | Description                 | Entry    | Notes |
|-----------|-----------------------------|----------|-------|
| CLSLOTS   | Number of CL Blocks         | 2        |       |
| NOPKG     | No. of Custom Data Packages | 2        |       |
| NUMSWTCH  | Number of Switches          | 0        |       |
| NOGINPTS  | Number of General Inputs    | 0        |       |
| NOGOPTS   | Number of General Outputs   | 0        |       |
| PKGNAME1  | Source File 1 Name          | B_PID3TM |       |
| PKGNAME2  | Source File 2 Name          | CLM_DMD3 |       |

*Continued on next page*

## 6.3.14 PID3TM, PID Controller, Continued

### Complete listing of CDS parameters

#### PID Controller Point CDS Parameters (B\_PID3TM)

| Parameter | ACCOL Signal | Notes                                                        |
|-----------|--------------|--------------------------------------------------------------|
| INPUT     | INPUT*       | Operator change writes from SP (mode dependent)              |
| SETPOINT  | SETPOINT*    |                                                              |
| DEADBAND  | DEADBAND     | Gain                                                         |
| K         | PROPORTIONAL | Repeats/Minute                                               |
| T1        | INTEGRAL     | Minutes                                                      |
| T2        | DERIVATIVE   | MAN output from TPS operator                                 |
| OP        | OP           | Read back from Analog Output (becomes TPS output on display) |
| CVIN      | RESET        | Input only                                                   |
| TRACK     | TRACK        |                                                              |
| SERROR    | ERROR        | Read only PID output (information only)                      |
| OUTPUT    | OUTPUT*      | Of analog output                                             |
| ZERO      | ZERO         | Of analog output                                             |
| SPAN      | SPAN         | User ACCOL signal (see below)                                |
| MODE      | AUTO         |                                                              |

\* If RBE is used, this signal should be configured as an RBE in the ACCOL build.

### ACCOL program

The user must provide a short ACCOL program to implement an external MODE signal for each PID. We recommend that the following be used:

If (Calculator. Auto)

\* Auto Mode

```
calculator
    pid.track = anot.track
    pid.reset = anot.reset
    calculator.op = ANOT.RESET
else
```

\* Manual Mode

```
calculator
    pid.track = #on
    pid.reset = calculator.op
endif
```

*Continued on next page*

## 6.3.14 PID3TM, PID Controller, Continued

### Builder input form

Bristol Babcock PID Builder Input Form (B\_PID3TM)

| Prompt   | Entry    | Description                                             |
|----------|----------|---------------------------------------------------------|
| ENTRY    | B_PID3TM | Template Name (.ZT)                                     |
| NAME     |          | TPS Point Name                                          |
| PIT_NAME |          | NDT PIT Name                                            |
| INPUT    |          | PID INPUT Signal Name                                   |
| SETPOINT |          | PID SETPOINT Signal Name                                |
| DEADBAND |          | PID DEADBAND Signal Name                                |
| K        |          | PID PROPORTIONAL Signal Name                            |
| T1       |          | PID INTEGRAL Signal Name                                |
| T2       |          | PID DERIVATIVE Signal Name                              |
| OP       |          | CALCULATOR OP Signal Name                               |
| RESET    |          | PID RESET Signal Name                                   |
| TRACK    |          | PID TRACK Signal Name                                   |
| SERROR   |          | PID ERROR Signal Name                                   |
| OUTPUT   |          | ANOUT OUTPUT Signal Name                                |
| ZERO     |          | ANOUT ZERO Signal Name                                  |
| SPAN     |          | ANOUT SPAN Signal Name                                  |
| AUTO     |          | CALCULATOR AUTO Signal Name                             |
| IO_SLOT1 |          | RTU slot number where I/O card with signal is installed |
| IO_SLOT2 |          | RTU slot number where I/O card with signal is installed |
| IO_SLOT3 |          | RTU slot number where I/O card with signal is installed |
| END      |          | End of current entry                                    |

*Continued on next page*

## 6.3.14 PID3TM, PID Controller, Continued

Builder input form,  
continued

|                  |
|------------------|
| <b>ATTENTION</b> |
|------------------|

---

The PID3TM image point requires that you link two CL blocks, named SPPID\_CL and CLPID\_CL. The SPPID\_CL block moves the setpoint signal as read back from the RTU to the SP parameter.

The second CL block performs the CL control algorithm function of computing the OP parameter.

These CL blocks are included in the package B\_PIDCL.AO. An unlink/link file is installed in your CL directory. To link or relink this block, the form of the command is

```
EC NET>UCLM>B_PIDCL.EC PAR1
```

where UCLM is the CL directory you named in the installation process, and PAR1 is the name of the point. An error may result if the modules are not already linked to the point.

---



## Section 7 – Smart Transmitter Configuration

### 7.1 Honeywell Smartline Transmitter Network Interface

- 
- Interface composition** The CLM interface to a Honeywell Smartline Transmitter within a Bristol Babcock network consists of the following:
- Smart Transmitter Configuration Display
  - STI Image Points with complete CDS (one for each smart transmitter)
- 

### 7.2 Bring Up the Smart Transmitter Configuration Display

| Action                          | Result                                         |
|---------------------------------|------------------------------------------------|
| Press SCHEM key                 | ENTER SCHEMATIC NAME message at top of display |
| Enter schematic name - CLMSMT00 | ENTER POINT ID message at top of display       |
| Enter POINT ID name             |                                                |

---

- Press assigned button on keyboard** A button can be assigned on the keyboard to bring up the smart transmitter configuration display, Reference 4.4.
- 

- Change point from Smart Transmitter Configuration Display** When the Smart Transmitter Configuration Display is on display, you may change points as follows:

#### Changing Points on Smart Transmitter Display

| Action                                                                                                                                                                                                                                               | Result                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| SELECT the <b>B_HWSTI</b> target (at top of Smart Transmitter Configuration Display).                                                                                                                                                                | ENTER POINT ID message appears at top of display.                                                          |
| Enter desired Point ID value.                                                                                                                                                                                                                        | The Smart Transmitter Configuration Display - 7.3 - appears with configuration and smart transmitter data. |
| <b>Note:</b> If the Point ID entered does not represent a smart transmitter point, the error message <code>INVALID DATA TYPE</code> will appear at the top of the display. You can then follow one of the steps above to enter the correct Point ID. |                                                                                                            |

---

## 7.3 Smart Transmitter Configuration Display

### Introduction

The Smart Transmitter Configuration Display is a custom display that emulates the PM's STI detail display, or CONFIG PAGE, both in appearance and operation. The display supports the configure, save, restore, rearrange, and calibration functions of the smart transmitter.

### Smart Transmitter Configuration Display

MAKE SELECTION12 Jul 98 09:34:29

B\_HWSTI    STT3000 TEST POINT    05 UNIT A5

CONFIG PAGE

----- CONFIGURATION DATA -----

|          |       |          |         |          |         |
|----------|-------|----------|---------|----------|---------|
| DEVICE   | 110   | PVFORMAT | D1      | PVEXEUI  | 5050.00 |
| CHANNEL  | 4     | PVSRCOPT | ALL     | PVEUHI   | 5000.00 |
| MISMATCH | ON    | PVCLAMP  | NOCLAMP | PVEULO   | 0.00    |
| COMERRS  | 0.00  | PVALDB   | HALF    | PVEXEULO | 0.00    |
| FREQ6050 | 60 HZ |          |         | DONE     | ON      |

CONFIGURATION STATUS: THIS IS THE XMITTER CONFIG

----- SMART TRANSMITTER DATA -----

|          |          |        |         |          |          |
|----------|----------|--------|---------|----------|----------|
| STITAG   | 12345678 | PVRAW  | 2500.00 | SECVAR   | 200.00   |
| SENSRTYP | STT      | URL    | 5000.00 | DAMPING  | 102.30   |
| PVCHAR   | LINEAR   | URV    | 4000.00 | SERIALNO | 5WY60102 |
| CJTACT   | OFF      | LRV    | 10.00   | STISWVER | 350R02   |
| PIUOTDCF | OFF      | LRL    | 0.00    | STATUS   | PROCMODE |
| DECONF   | PV_SV    | STI_EU | OHMS    | COMMAND  | PROCMODE |

TRANSMITTER SCRATCH PAD: - - - SCRATCH PAD DATA - - -

TRANSMITTER STATUS            : - - - XMITTER STATUS INFORMATION -

6189

*Continued on next page*

## 7.3 Smart Transmitter Configuration Display, Continued

### Smart transmitter display contents

The Smart Transmitter Configuration Display consists of the following four general areas:

- Command line, date/time, and heading line at top of display
- Configuration Data
  - Parameters and values
  - Configuration status line
- Smart Transmitter Data
  - Parameters and values
  - Transmitter scratch pad and status lines
- Variable Target Area
  - Variable targets display on left side, depending on parameter selected
  - Selection of any one of the following parameters will display appropriate targets at the left side of the display.

#### Configuration Data

FREQ6050

#### Smart Transmitter Data

|          |       |           |
|----------|-------|-----------|
| PVCHAR   | - URV | - DAMPING |
| CJTACT   | - LRV |           |
| PIUOTDCF |       |           |
| DECONF   |       | - COMMAND |

## 7.4 Smart Transmitter Display After DECONF Selection

### Smart transmitter display after DECONF selection

Smart Transmitter Configuration display, after selection of the DECONF parameter. Note the additional information displayed at the left side of the display.

- Parameter name selected displays at top of target area (in this example, "DECONF").
- Targets corresponding to alternative values for the parameter selected (in this example, four targets representing four possible value selections to configure for the "DECONF" parameter).
- ENTER target at bottom of target area: Select the desired value target among alternatives displayed, then ENTER target to configure parameter to selected value. **Note:** To use the display you must select either PV-DB or PV-SV-DB.

### Smart Transmitter Configuration Display (After DECONF Selection)

MAKE SELECTION12 Jul 98 09:34:38

B\_HWSTI STT3000 TEST POINT 05 UNIT A5

CONFIG PAGE

DECONF

PV

PV\_SV

PV\_DB

PV\_SV\_DB

----- CONFIGURATION DATA -----

|          |       |          |         |          |         |
|----------|-------|----------|---------|----------|---------|
| DEVICE   | 110   | PVFORMAT | D1      | PVEXEUI  | 5050.00 |
| CHANNEL  | 4     | PVSRCOPT | ALL     | PVEUHI   | 5000.00 |
| MISMATCH | ON    | PVCLAMP  | NOCLAMP | PVEULO   | 0.00    |
| COMERRS  | 0.00  | PVALDB   | HALF    | PVEXEULO | 0.00    |
| FREQ6050 | 60 HZ |          |         | DONE     | ON      |

CONFIGURATION STATUS: THIS IS THE XMITTER CONFIG

----- SMART TRANSMITTER DATA -----

|          |          |        |         |          |           |
|----------|----------|--------|---------|----------|-----------|
| STITAG   | 12345678 | PVRAW  | 2500.00 | SECVAR   | 200.00    |
| SENSRTYP | STT      | URL    | 5000.00 | DAMPING  | 102.30    |
| PVCHAR   | LINEAR   | URV    | 4000.00 | SERIALNO | 5WY60102  |
| CJTACT   | OFF      | LRV    | 10.00   | STISWVER | 350R02    |
| PIUOTDCF | OFF      | LRL    | 0.00    | STATUS   | PROC MODE |
| DECONF   | PV_SV    | STI_EU | OHMS    | COMMAND  | PROC MODE |

TRANSMITTER SCRATCH PAD: - - - SCRATCH PAD DATA - - -

TRANSMITTER STATUS : - - - XMITTER STATUS INFORMATION -

ENTER

6190

## 7.5 Configuration Data of Smart Transmitter Display

### Smart Transmitter display parameters

Description of each parameter in the Configuration Data section of the Smart Transmitter Configuration display. Parameters are documented from left to right.

#### Configuration Data Parameters

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MISMATCH  | <p>Mismatch Code: If = ON, indicates at least one configuration parameter in the smart transmitter does not match the signal value representing the parameter. If = OFF, no parameters mismatch. (Invalid configuration parameters are treated as an implicit mismatch, since it is not possible for the transmitter to have an invalid parameter value.)</p> <p>Mismatch detection occurs only if the transmitter's DECONF parameter is equal to PV_DB or PV_SV_DB. A mismatch condition is assumed when the system first powers up, and when an SFC writes to a parameter that affects the PV. The Questionable Data status is set for all signals with a mismatch condition. The mismatch condition can be cleared by a Download or Upload command, or when the transmitter sends its database and all parameters are compared equal.</p> <p>The following numeric signals are checked for a mismatch with the transmitter. The string and logical signals (*) are not checked.</p> <p>SENSRTYP<br/>DECONF<br/>DAMPING<br/>PVCHAR<br/>CJTACT*<br/>PIUOTDCF*<br/>STITAG*<br/>FREQ6050*<br/>LRV<br/>URV<br/>URL</p> |
| COMERRS   | Communication Error Count: Displays a count of transmitter communication errors detected. The counter is modulo 256.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| FREQ6050  | Frequency Filter Code: Indicates the smart temperature transmitter line frequency filter used: Valid values are 50 Hz or 60 Hz.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PVFORMAT  | <p>PV Decimal Point Format Code: Defines the decimal format used to display the PV and SP values. Following are valid code values</p> <p>D0 ..... No decimals<br/>D1 ..... One decimal<br/>D2 ..... Two decimals<br/>D3 ..... Three decimals</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

*Continued on next page*

## 7.5 Configuration Data of Smart Transmitter Display,

Continued

Smart Transmitter  
configuration display,

| Parameter            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PVSRCOPT             | PV Source Option: Defines the PV source option for this data point, per the following:<br>ALL ..... PV provided by operator, sequence program, or by field wiring.<br>ONLYAUTO ..... PV source selection is not available.                                                                                                                                                                                                                                                                                                                                                |
| PVCLAMP              | PV Clamping Option: Defines whether PV clamping is used for this data point. If = CLAMP and the PV extended range is exceeded; PV value status (PVSTS) is marked Uncertain and the PV is set equal to the extended limit that was violated. Valid options are:<br>NOCLAMP ..... No clamping of the PV value<br>CLAMP ..... Clamp PV value at range extension limit                                                                                                                                                                                                        |
| PVALDB               | PV Alarm Deadband: Alarm deadband is used to prevent excessive recurrence of alarms by adjusting the percent of the Engineering Unit (EU) range at which the alarm "returns to normal." An increase in the deadband percent of PV range (PVEULO to PVEUHI) causes PV alarms to return to normal less frequently. Valid options are:<br>HALF ..... 1/2 of 1% of EU range<br>ONE ..... 1% of EU range<br>TWO ..... 2% of EU range<br>THREE ..... 3% of EU range<br>FOUR ..... 4% of EU range<br>FIVE ..... 5% of EU range<br>EU ..... Specify deadband in Engineering Units |
| PVEXEUHI             | PV Extended Engineering Unit Range High: Limit value used to clamp or detect a bad PV value; must be => PVEUHI, NaN. <b>Note:</b> PVEXEUHI cannot be written with NaN; NaN is the default value only.                                                                                                                                                                                                                                                                                                                                                                     |
| PVEUHI               | PV High Range in Engineering Units: Valid range is PVEULO to PVEXEUHI, NaN. <b>Note:</b> PVEUHI cannot be written with NaN; NaN is the default value only.                                                                                                                                                                                                                                                                                                                                                                                                                |
| PVEULO               | PV Low Range in Engineering Units: Valid range is PVEXEULO to PVEUHI, NaN. <b>Note:</b> PVEULO cannot be written with NaN; NaN is the default value only.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PVEXEULO             | PV Extended Engineering Unit Low Range: Limit value used to clamp or detect a bad PV value; must be =< PVEULO, NaN. <b>Note:</b> PVEXEULO cannot be written with NaN; NaN is the default value only.                                                                                                                                                                                                                                                                                                                                                                      |
| DONE                 | Done Flag: Flag is used with the eight transmitter COMMANDs to indicate completion of a transmitter command: Set to OFF at the beginning of a transmitter command and set to ON when the command is completed.                                                                                                                                                                                                                                                                                                                                                            |
| CONFIGURATION STATUS | Transmitter Configuration Status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 7.6 Parameter Description of Smart Transmitter Display

### Introduction

Description of each parameter in the Smart Transmitter Data section of the Smart Transmitter Configuration display. Parameters are documented from left to right.

#### Smart Transmitter Data Parameters

| Parameter     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------|------------------------------|---------|--------|-----------------------|--------|--------|--------------------------|--------|--------|---------------------|--------|---------|--------------------------|----------|----------|----------|--------|-------|---------|----------|---------|--|
| STITAG        | Transmitter Tag Name: Tag name is used to identify a transmitter to the system and on displays, reports, and logs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| SENSRTYP      | <div>Sensor Type: Defines the Smart Transmitter type. This is a read/write configuration parameter. Transmitter type codes are:</div> <table><tr><td>8</td><td>SPT_DP</td><td>ST3000 differential pressure</td></tr><tr><td>9</td><td>SPT_GP</td><td>ST3000 gauge pressure</td></tr><tr><td>10</td><td>SPT_AP</td><td>ST3000 absolute pressure</td></tr><tr><td>11</td><td>STT</td><td>STT3000 temperature</td></tr><tr><td>12</td><td>SFM</td><td>MagneW3000 magnetic flow</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8                            | SPT_DP | ST3000 differential pressure | 9       | SPT_GP | ST3000 gauge pressure | 10     | SPT_AP | ST3000 absolute pressure | 11     | STT    | STT3000 temperature | 12     | SFM     | MagneW3000 magnetic flow |          |          |          |        |       |         |          |         |  |
| 8             | SPT_DP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ST3000 differential pressure |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| 9             | SPT_GP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ST3000 gauge pressure        |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| 10            | SPT_AP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ST3000 absolute pressure     |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| 11            | STT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | STT3000 temperature          |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| 12            | SFM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MagneW3000 magnetic flow     |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| PVCHAR        | <div>PV Characterization Option: Defines the display characterization to be used for characterizing the input PV value. Valid values for PVCHAR are based on the transmitter type (SENSRTYP), as follows:</div> <div>For differential pressure transmitter SPT_DP<br/>LINEAR<br/>SQRROOT</div> <div>For pressure transmitters SPT_GP and SPT_AP<br/>LINEAR</div> <div>For temperature transmitter STT</div> <table><tr><th>Thermocouples</th><th>RTDs</th></tr><tr><td>BTHERM</td><td>CU10RTD</td></tr><tr><td>ETHERM</td><td>CU25RTD</td></tr><tr><td>JTHERM</td><td></td></tr><tr><td>KTHERM</td><td>DINRTD</td></tr><tr><td>RTHERM</td><td>JISRTD</td></tr><tr><td>STHERM</td><td>NICKLRD</td></tr><tr><td>TTHERM</td><td>PT200RTD</td></tr><tr><td>NINIMOTC</td><td>PT500RTD</td></tr><tr><td>NTHERM</td><td>RHRAD</td></tr><tr><td>W3W25TC</td><td>RTD_OHMS</td></tr><tr><td>W5W26TC</td><td></td></tr></table> <div>For magnetic flow transmitter SFM<br/>LINEAR</div> | Thermocouples                | RTDs   | BTHERM                       | CU10RTD | ETHERM | CU25RTD               | JTHERM |        | KTHERM                   | DINRTD | RTHERM | JISRTD              | STHERM | NICKLRD | TTHERM                   | PT200RTD | NINIMOTC | PT500RTD | NTHERM | RHRAD | W3W25TC | RTD_OHMS | W5W26TC |  |
| Thermocouples | RTDs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| BTHERM        | CU10RTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| ETHERM        | CU25RTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| JTHERM        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| KTHERM        | DINRTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| RTHERM        | JISRTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| STHERM        | NICKLRD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| TTHERM        | PT200RTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| NINIMOTC      | PT500RTD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| NTHERM        | RHRAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| W3W25TC       | RTD_OHMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| W5W26TC       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |
| CJFACT        | Internal Cold Junction Temperature: Defines whether the Smart Temperature Transmitter's internal cold-junction reference is used (ON), or an externally-provided cold-junction reference is used (OFF).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                              |        |                              |         |        |                       |        |        |                          |        |        |                     |        |         |                          |          |          |          |        |       |         |          |         |  |

*Continued on next page*

## 7.6 Parameter Description of Smart Transmitter Display,

Continued

Smart Transmitter Data Parameters, Continued

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PIUOTDCF  | Open Thermocouple Detection Enable: Defines whether the point is to detect an open thermocouple condition (ON), or not to detect an open thermocouple (OFF). This parameter is configurable for each STI point that is connected to a smart temperature transmitter.                                                                                                                 |
| DECONF    | Digitally Enhanced Configuration Mode: Defines the contents of the data that will be sent by the smart transmitter to the STI point. Following are the valid Bristol Babcock DE Configuration Mode signal codes:<br><br>PV ..... PV only<br>PV_SV ..... PV and Secondary Variable (SV)<br>PV_DB ..... PV and transmitter database<br>PV_SV_DB ..... PV, SV, and transmitter database |
| PVRAW     | PV value: Indicates the value of the PV calculated in the RTU based on the transmitter PC, after PV characterization (PVCHAR) and DAMPING have been performed. (The span of the PV is determined by using LRV as a 0%- point and URV as a 100%-point.)                                                                                                                               |
| URL       | Upper Range Limit: Indicates the upper range limit of the PV at the smart transmitter. This limit is a fixed limit and cannot be changed. Refer to the description of the STI_EU parameter for the URL engineering units.                                                                                                                                                            |
| URV       | Upper Range Value: Defines the upper end of the operating range for the PV parameter value. Refer to the description of the STI_EU parameter for the URV engineering units. User entry for PVEUHI is the user-entered engineering-unit value that corresponds to URV. Valid values for URV must be in the range: LRV =< (URV VALUE) =< URL. A value of NaN is invalid.               |
| LRV       | Lower Range Value: Defines the lower end of the operating range for the PV parameter value. Refer to the description of the STI_EU parameter for the LRV engineering units. User entry for PVEULO is the user-entered engineering-unit value that corresponds to LRV. Valid values for LRV must be in the range: LRL => (LRV VALUE) =< URV. A value of NaN is invalid.               |
| LRL       | Lower Range Limit: Indicates the lower range limit of the PV at the smart transmitter. This limit is a fixed limit and cannot be changed. Refer to the description of the STI_EU parameter for the LRL engineering units.                                                                                                                                                            |

*Continued on next page*

## 7.6 Parameter Description of Smart Transmitter Display, Continued

### Smart Transmitter Data Parameters, Continued

| Parameter | Description                                                                                                                                                                          |                         |                                 |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------|
| STI_EU    | Smart Transmitter Engineering Units: Defines the units of measurement for following parameters: LRL, LRV, URL, and URV. Following are the valid Bristol Babcock STI_EU signal codes: |                         |                                 |
|           | <b>Code</b>                                                                                                                                                                          | <b>Transmitter Type</b> | <b>Description</b>              |
|           | 0 - INH2O                                                                                                                                                                            | Pressure                | Inches of water                 |
|           | 1 - MMHG                                                                                                                                                                             | Pressure                | Millimeters of mercury          |
|           | 2 - PSI                                                                                                                                                                              | Pressure                | Pounds per square inch          |
|           | 3 - KPA                                                                                                                                                                              | Pressure                | Kilopascals                     |
|           | 4 - MPA                                                                                                                                                                              | Pressure                | Millipascals                    |
|           | 5 - MBAR                                                                                                                                                                             | Pressure                | Millibars                       |
|           | 6 - BAR                                                                                                                                                                              | Pressure                | Bars                            |
|           | 7 - G_SQCM                                                                                                                                                                           | Pressure                | Grams per square centimeter     |
|           | 8 - KG_SQCM                                                                                                                                                                          | Pressure                | Kilograms per square centimeter |
|           | 9 - MMH2O                                                                                                                                                                            | Pressure                | Millimeters of water            |
|           | 10 - INHG                                                                                                                                                                            | Pressure                | Inches of mercury               |
|           | 11 - DEG_C                                                                                                                                                                           | Temperature             | Degrees Centigrade              |
|           | 12 - DEG_F                                                                                                                                                                           | Temperature             | Degrees Fahrenheit              |
|           | 13 - DEG_K                                                                                                                                                                           | Temperature             | Degrees Kelvin                  |
|           | 14 - DEG_R                                                                                                                                                                           | Temperature             | Degrees Rankine                 |
|           | 15 - MV                                                                                                                                                                              | Temperature             | Millivolts                      |
|           | 16 - V                                                                                                                                                                               | Temperature             | Volts                           |
|           | 17 - OHMS                                                                                                                                                                            | Temperature             | Ohms                            |
|           | 18 - CM_HR                                                                                                                                                                           | Magnetic flow           | Cubic meters per hour           |
|           | 19 - GAL_HR                                                                                                                                                                          | Magnetic flow           | Gallons per hour                |
|           | 20 - LIT_HR                                                                                                                                                                          | Magnetic flow           | Liters per hour                 |
|           | 21 - CC_HR                                                                                                                                                                           | Magnetic flow           | Cubic centimeters per hour      |
|           | 22 - CM_MIN                                                                                                                                                                          | Magnetic flow           | Cubic meters per minute         |
|           | 23 - GAL_MIN                                                                                                                                                                         | Magnetic flow           | Gallons per minute              |
|           | 24 - LIT_MIN                                                                                                                                                                         | Magnetic flow           | Liters per minute               |
|           | 25 - CC_MIN                                                                                                                                                                          | Magnetic flow           | Cubic centimeters per minute    |
|           | 26 - CM_DAY                                                                                                                                                                          | Magnetic flow           | Cubic meters per day            |
|           | 27 - GAL_DAY                                                                                                                                                                         | Magnetic flow           | Gallons per day                 |
|           | 28 - KGAL_DAY                                                                                                                                                                        | Magnetic flow           | Kilogallons per day             |
|           | 29 - BRL_DAY                                                                                                                                                                         | Magnetic flow           | Barrels per day                 |
|           | 30 - CM_SEC                                                                                                                                                                          | Magnetic flow           | Cubic meters per second         |
|           | <b>Note:</b> STI_EU may not be configured or changed from the CLM. It may be configured in the ACCOL load or changed using the PC Tool kit function.                                 |                         |                                 |
|           | <b>Note:</b> If the STI_EU value is invalid or not used, STI_EU defaults to the transmitter type base units, provided that SENSRTYP is valid, per the following defaults:            |                         |                                 |
|           | Pressure transmitter ..... 0 - INH2O (Inches of water)                                                                                                                               |                         |                                 |
|           | Temperature transmitter ..... 11 - DEG_C (Degrees Centigrade)                                                                                                                        |                         |                                 |
|           | Flow transmitter ..... 18 - CM_HR (Cubic meters per hour)                                                                                                                            |                         |                                 |

*Continued on next page*

## 7.6 Parameter Description of Smart Transmitter Display, Continued

Smart Transmitter Data Parameters, Continued

| Parameter      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|------------|-----------|-----------|------|------------|-----------|-----|------------|-----------|-----|------------|-----------|-----|-----------|-----------|-----|-----------|-----------|-----|-----------|------------|-----|-----------|------------|------|------------|------------|------|------------|-------------|-------|
| SECVAR         | Secondary Variable: Displays the value of the secondary variable of the smart transmitter as follows:<br><br>Pressure transmitter ..... Temperature of the transmitter<br>Temperature transmitter ..... Cold junction temperature<br>Flow transmitter ..... Totalized value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| DAMPING        | Damping: Defines the first-order PV filtering option for the smart transmitter. When a Download command is issued, this parameter is adjusted to the nearest appropriate value based on the transmitter type. At the completion of a download, the signal value is set to the adjusted value. The damping constant selected is based on the DAMPING parameter value and the transmitter type specified with the SENSRTYP parameter. DAMPING constants follow:<br><br><b>Transmitter Type</b><br><br><table><tr><th>Pressure (SPT)</th><th>Temperature (STT)</th><th>Flow (SFM)</th></tr><tr><td>0.0 .....</td><td>0.0 .....</td><td>0.00</td></tr><tr><td>0.16 .....</td><td>0.3 .....</td><td>0.5</td></tr><tr><td>0.32 .....</td><td>0.7 .....</td><td>1.0</td></tr><tr><td>0.48 .....</td><td>1.5 .....</td><td>2.0</td></tr><tr><td>1.0 .....</td><td>3.1 .....</td><td>3.0</td></tr><tr><td>2.0 .....</td><td>6.3 .....</td><td>4.0</td></tr><tr><td>4.0 .....</td><td>12.7 .....</td><td>5.0</td></tr><tr><td>8.0 .....</td><td>25.5 .....</td><td>10.0</td></tr><tr><td>16.0 .....</td><td>51.1 .....</td><td>50.0</td></tr><tr><td>32.0 .....</td><td>102.3 .....</td><td>100.0</td></tr></table> | Pressure (SPT) | Temperature (STT) | Flow (SFM) | 0.0 ..... | 0.0 ..... | 0.00 | 0.16 ..... | 0.3 ..... | 0.5 | 0.32 ..... | 0.7 ..... | 1.0 | 0.48 ..... | 1.5 ..... | 2.0 | 1.0 ..... | 3.1 ..... | 3.0 | 2.0 ..... | 6.3 ..... | 4.0 | 4.0 ..... | 12.7 ..... | 5.0 | 8.0 ..... | 25.5 ..... | 10.0 | 16.0 ..... | 51.1 ..... | 50.0 | 32.0 ..... | 102.3 ..... | 100.0 |
| Pressure (SPT) | Temperature (STT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Flow (SFM)     |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 0.0 .....      | 0.0 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.00           |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 0.16 .....     | 0.3 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.5            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 0.32 .....     | 0.7 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1.0            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 0.48 .....     | 1.5 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.0            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 1.0 .....      | 3.1 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3.0            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 2.0 .....      | 6.3 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4.0            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 4.0 .....      | 12.7 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.0            |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 8.0 .....      | 25.5 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10.0           |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 16.0 .....     | 51.1 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 50.0           |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| 32.0 .....     | 102.3 .....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 100.0          |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| SERIALNO       | Serial Number: Displays the serial number of the smart transmitter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |
| STISWVER       | Software Revision Level: Displays the software revision level of the smart transmitter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                   |            |           |           |      |            |           |     |            |           |     |            |           |     |           |           |     |           |           |     |           |            |     |           |            |      |            |            |      |            |             |       |

*Continued on next page*

## 7.6 Parameter Description of Smart Transmitter Display,

Continued

Smart Transmitter Data Parameters, Continued

| Parameter | Description                                                                                                                                            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATUS    | Status Code: Indicates the current state of the HWSTI module, including the operating mode, command status, and error conditions. Valid status codes : |
| 0         | PROCMode ..... Process mode                                                                                                                            |
| 1         | PROCMode ..... Process mode with parameter(s) mismatch                                                                                                 |
| 2         | CONFMode ..... Configure mode                                                                                                                          |
| 3         | DNLDEROR ..... Download completed with rejected parameter                                                                                              |
| 4         | DOWNLOAD ..... Download command in progress                                                                                                            |
| 5         | DNLD OK ..... Download completed OK                                                                                                                    |
| 6         | DNLDFAIL ..... Download failed                                                                                                                         |
| 7         | UPLOAD ..... Upload command in progress                                                                                                                |
| 8         | LOADCOMP ..... Upload completed OK                                                                                                                     |
| 9         | LOADFAIL ..... Upload failed                                                                                                                           |
| 10        | SET_LRV ..... Set Lower Range command in progress                                                                                                      |
| 11        | S_LRV OK ..... Set Lower Range completed OK                                                                                                            |
| 12        | SLRVFAIL ..... Set Lower Range failed                                                                                                                  |
| 13        | SET_URV ..... Set Upper Range command in progress                                                                                                      |
| 14        | S_URV OK ..... Set Upper Range completed OK                                                                                                            |
| 15        | SURVFAIL ..... Set Upper Range failed                                                                                                                  |
| 16        | COR_LRV ..... Correct Lower Range command in progress                                                                                                  |
| 17        | C_LRV OK ..... Correct Lower Range completed OK                                                                                                        |
| 18        | CLRVFAIL ..... Correct Lower Range failed                                                                                                              |
| 19        | COR_URV ..... Correct Upper Range command in progress                                                                                                  |
| 20        | C_URV OK ..... Correct Upper Range completed OK                                                                                                        |
| 21        | CURVFAIL ..... Correct Upper Range failed                                                                                                              |
| 22        | COR_ZERO ..... Correct Zero Point command in progress                                                                                                  |
| 23        | CZERO OK ..... Correct Zero Point completed OK                                                                                                         |
| 24        | ZEROFAIL ..... Correct Zero Point failed                                                                                                               |
| 25        | CALIB ..... Set Calibration Defaults command in progress                                                                                               |
| 26        | CALCOMP ..... Set Calibration Defaults completed OK                                                                                                    |
| 27        | CALFAIL ..... Set Calibration Defaults failed                                                                                                          |
| 28        | INVALID ..... Invalid COMMAND value                                                                                                                    |
| 29        | POWRFAIL ..... Command failure due to a power failure                                                                                                  |
| 30        | PVTIMEOUT ..... Transmitter is not operating (PV timeout)                                                                                              |
| 31        | BORDFAIL ..... Interface (I/O board) hardware failure                                                                                                  |
| 32        | INVLDDDEV ..... Invalid DEVICE and/or CHANNEL specified                                                                                                |
| 99        | TESTMODE ..... Test mode, imaging transmitter data                                                                                                     |

*Continued on next page*

## 7.6 Parameter Description of Smart Transmitter Display, Continued

Smart Transmitter Data Parameters, Continued

| Parameter               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMMAND                 | <p>Command: Specifies the Bristol software module's operating mode, and allows the user to do database transfers between the image point and the smart transmitter, and also calibrate the transmitter. Following are valid signal codes:</p> <p>PROCMode ..... Process mode, Inhibit configuration signals<br/> CONFMode ..... Configure mode, Enable configuration signals<br/> TESTMode ..... Test mode</p> <p>Transmitter CONFMode commands</p> <p>DNLOADDB ..... Download transmitter parameters, from point into transmitter<br/> UPLOADDB ..... Upload transmitter parameters, from transmitter into point<br/> SET_LRV ..... Set the Lower Range Value (LRV)<br/> SET_URV ..... Set the Upper Range Value (URV)<br/> COR_LRV ..... Correct the Lower Range Value (LRV)<br/> COR_URV ..... Correct the Upper Range Value (URV)<br/> COR_INPT ..... Correct the zero point for the PV value<br/> RST_COR ..... Set all input calibration parameters to their default values</p> <p>Notes regarding Transmitter commands:</p> <ul style="list-style-type: none"> <li>• When transmitter command is specified, the PV and SECVAR signal values are set to NaN, and Questionable Data status is set for both signals.</li> <li>• While a command is in progress, changes to COMMAND value are ignored.</li> <li>• At completion of a command, the same command cannot be issued (until there is a change in COMMAND value).</li> <li>• If same command is desired, change the COMMAND value to Configure Mode (CONFMode), then back to the command value.</li> <li>• Parameter values are checked for mismatch at the completion of a command.</li> </ul> |
| TRANSMITTER SCRATCH PAD | Transmitter Scratch Pad: Displays the contents of the smart transmitter's scratch pad. It is equivalent to the first part of the S1 parameter in the TPS System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| TRANSMITTER STATUS      | Transmitter Status: Displays the contents of the smart transmitter's detailed status. It is equivalent to the second part of the S1 parameter in a TPS System.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Section 8 – Terminology and Definitions

### 8.1 Terms Used in This Document

---

#### Terms Used

**CLM** - A CLM is defined as a TPS Application Module (AM) with at least one CLMI or CCP communication board and the CLM software package installed. The AM must be an R300 (or later) revision with at least 4 megawords of memory.

**Image Point** - An operating multi-parameter point in the AM/CLM matching an ACCOL module in the RTU.

**Network Device** - refers to the non-Honeywell device, RTU, or network that interfaces to the TPS system by way of the CLM.

**Network Memory** - the portion of CLM memory defined for shared access with the CLMI/CCP board to perform the network communication task.

---

### 8.2 CLM/Bristol Babcock RTU Interface Package

#### RTU interface package items

The CLM/Bristol Babcock RTU Interface Package includes the following items:

- *CLM/Bristol Babcock RTU Interface* manual (this document)
  - *CLM/Bristol Babcock Parameter Reference Dictionary*
  - Software files supplied on loadable media - Ref. 3.5.
- 

### 8.3 Other Reference Documents

---

#### Reference documents

Bristol Babcock Documentation:

- *ACCOL II Reference for PROM Set AD and ACCOL Software Revision 5.3*
- *NETWORK 3000 Communication User's Guide #D4052*
- *Network Topology (NETTOP) Program User Manual #D4057*

AGA reference documents are listed below:

- *NETWORK 3000 Communication User's Guide #D4052*
  - *AGA Report 5, The Method Used to Calculate Fuel Gas Energy*
  - *AGA Report 7, "Measurement of Fuel Gas by Turbine Meters"*
-



## Appendix – UCLM: CCP AND UCIM Redesigned

CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

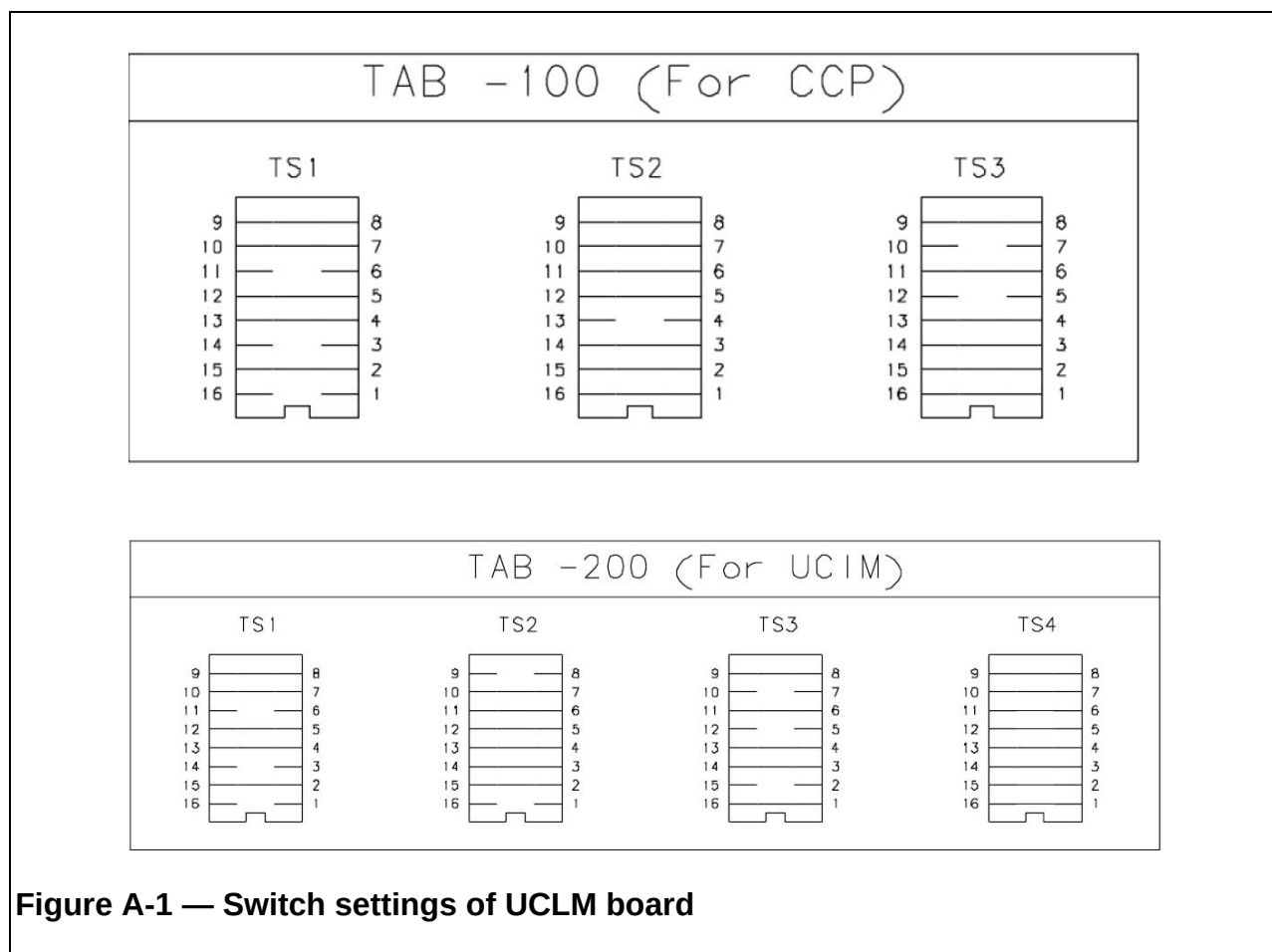
### A.1 ORDERING INFORMATION

To order the boards use the Honeywell part numbers as provided in the following table:

| Part Number  | Board Name |
|--------------|------------|
| 51306409-100 | CCP        |
| 51306409-200 | UCIM       |

### A.2 SWITCH SETTINGS OF UCLM BOARD

Figure A-1 illustrates the redesigned CCP and UCIM settings.



**Figure A-1 — Switch settings of UCLM board**

There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM





**Honeywell**

---

Honeywell International  
Process Solutions  
2500 West Union Hills  
Phoenix, AZ 85027